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Wu et al.

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(54) **SEMICONDUCTOR DEVICES**

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H01L 21/82 (2006.01)
H01L 21/768 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **H10D 84/038** (2025.01); **H01L 21/76805** (2013.01); **H01L 21/7682** (2013.01);
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(58) **Field of Classification Search**
CPC H10D 84/038; H10D 64/017; H10D 84/0135; H10D 84/0149; H10D 84/0158;
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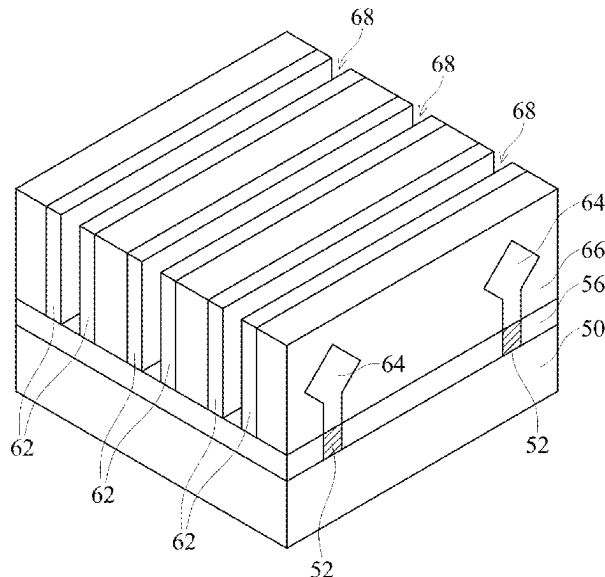
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(57) **ABSTRACT**
In an embodiment, a method includes: forming a first fin extending from a substrate; forming a second fin extending from the substrate, the second fin being spaced apart from the first fin by a first distance; forming a metal gate stack over the first fin and the second fin; depositing a first inter-layer dielectric over the metal gate stack; and forming a gate contact extending through the first inter-layer dielectric to physically contact the metal gate stack, the gate contact being laterally disposed between the first fin and the second fin, the gate contact being spaced apart from the first fin by a second distance, where the second distance is less than a second predetermined threshold when the first distance is greater than or equal to a first predetermined threshold.

20 Claims, 20 Drawing Sheets



Related U.S. Application Data

continuation of application No. 17/862,628, filed on Jul. 12, 2022, now Pat. No. 11,721,590, which is a continuation of application No. 17/169,809, filed on Feb. 8, 2021, now Pat. No. 11,393,724, which is a continuation of application No. 16/567,053, filed on Sep. 11, 2019, now Pat. No. 10,957,604.

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H10D 64/01 (2025.01)

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H10D 84/03 (2025.01)

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(58) **Field of Classification Search**

CPC H10D 30/0215; H10D 30/6215; H10D 30/024; H01L 21/76805; H01L 21/7682; H01L 21/76895; H01L 21/76816; H01L 21/027

USPC 438/283

See application file for complete search history.

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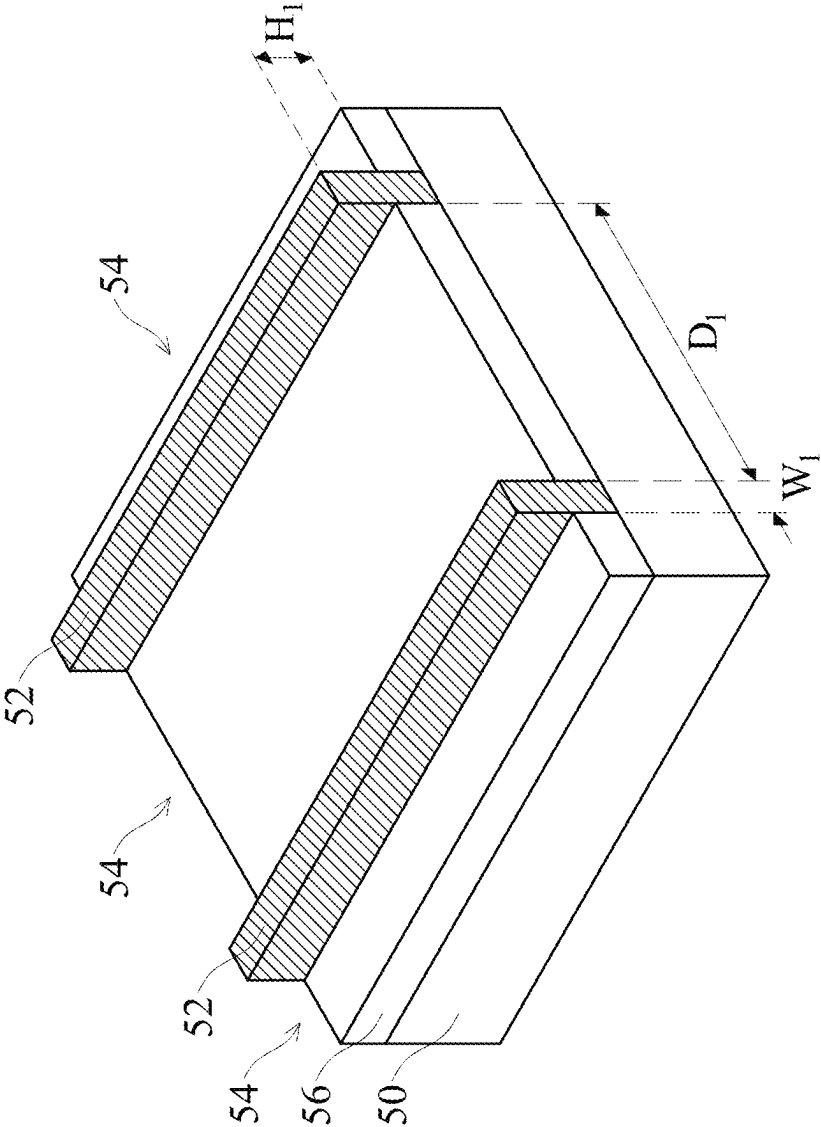


FIG. 1A

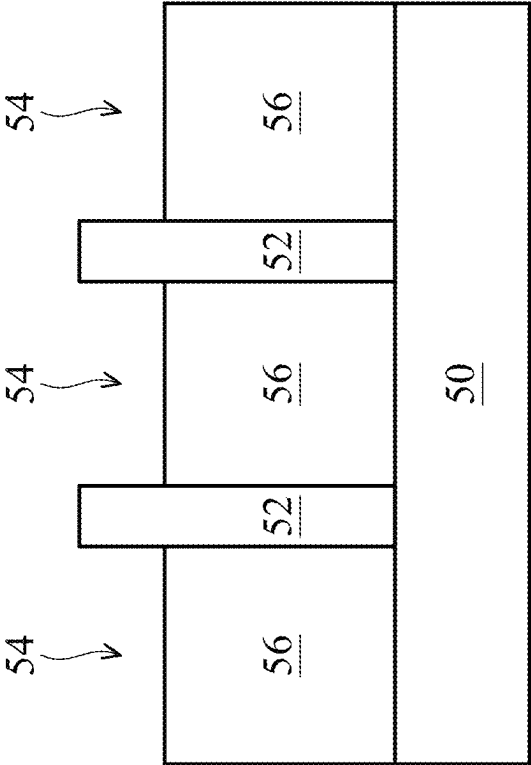


FIG. 1B

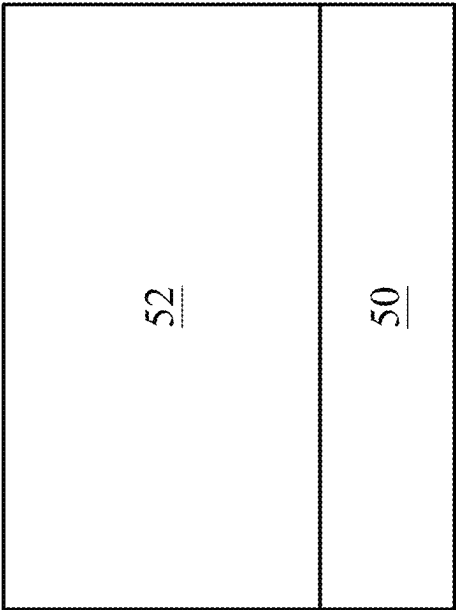


FIG. 1C

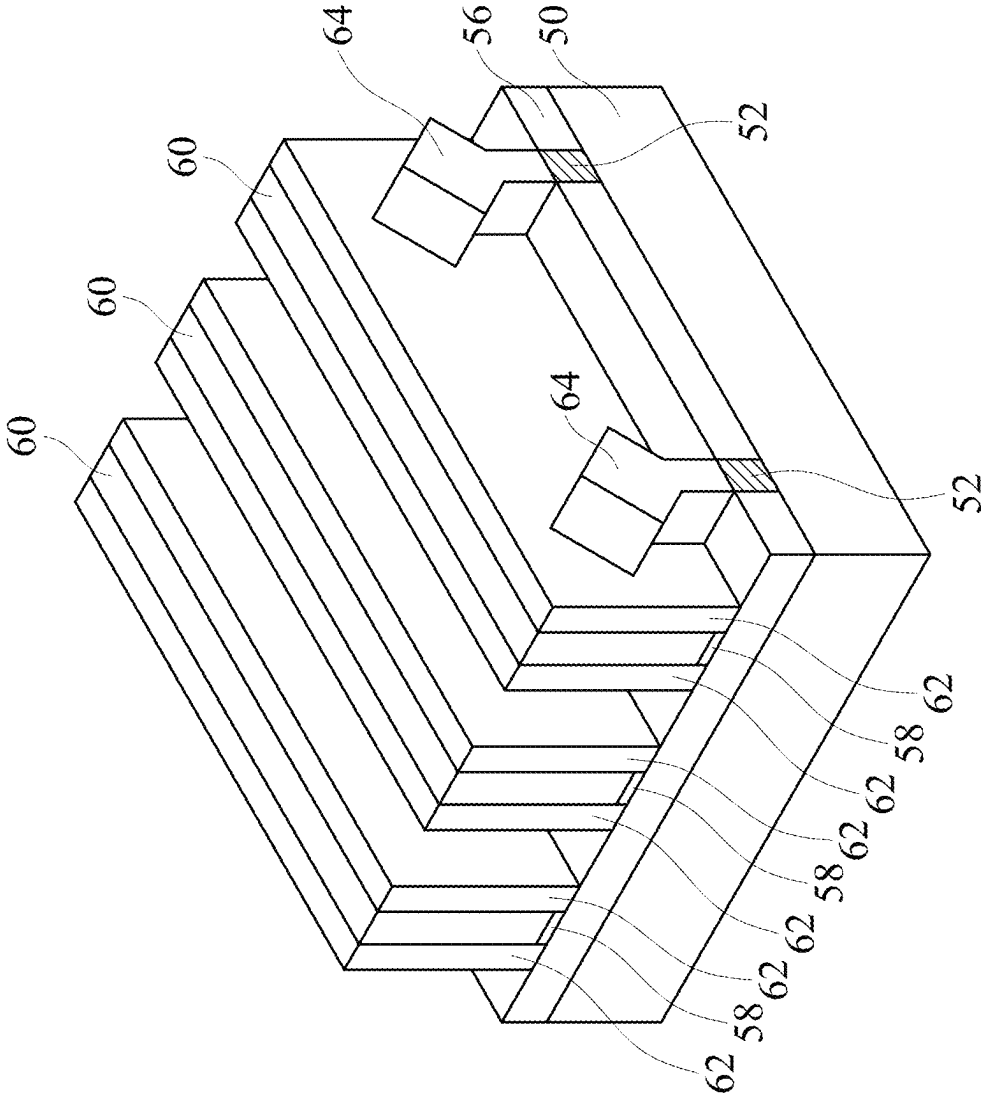


FIG. 2A

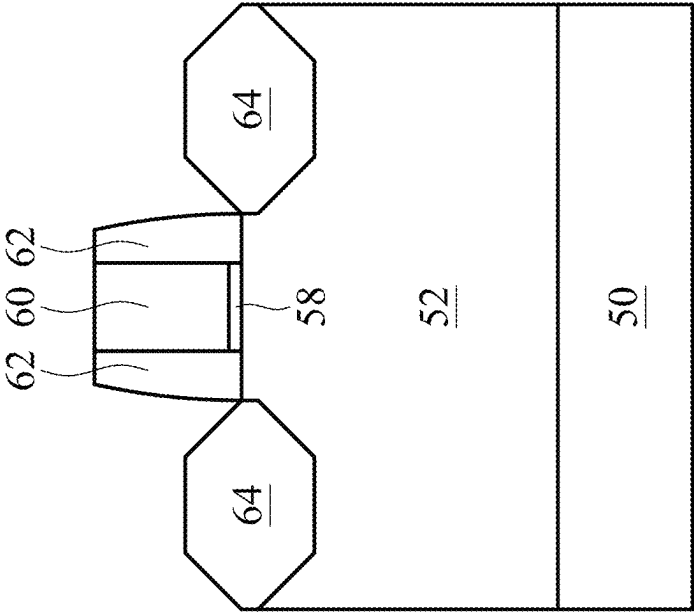


FIG. 2C

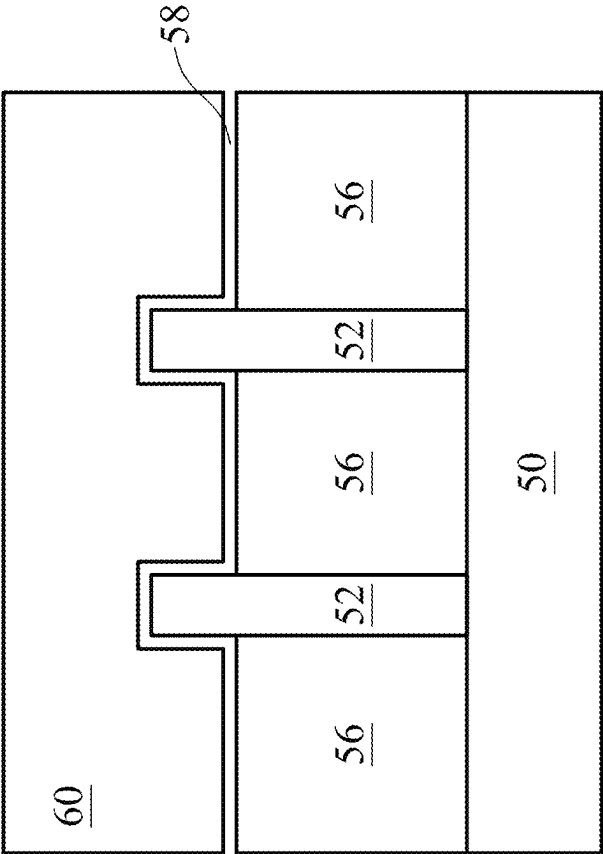


FIG. 2B

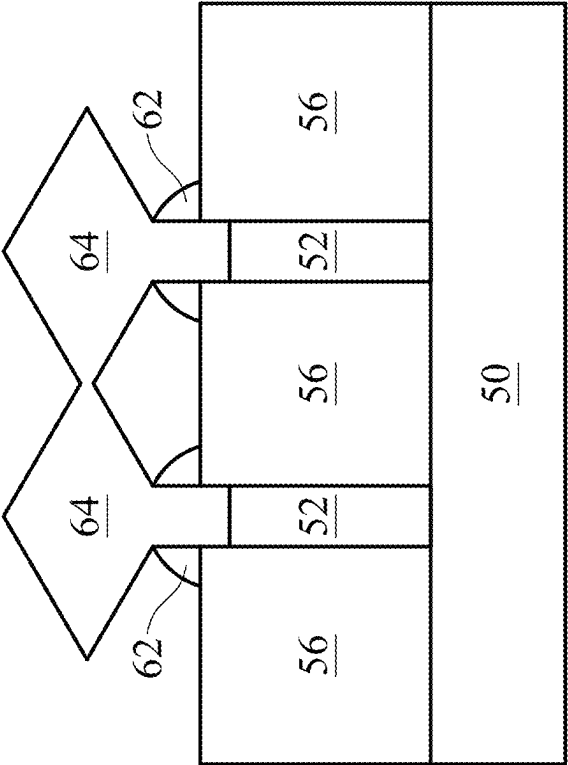


FIG. 2D

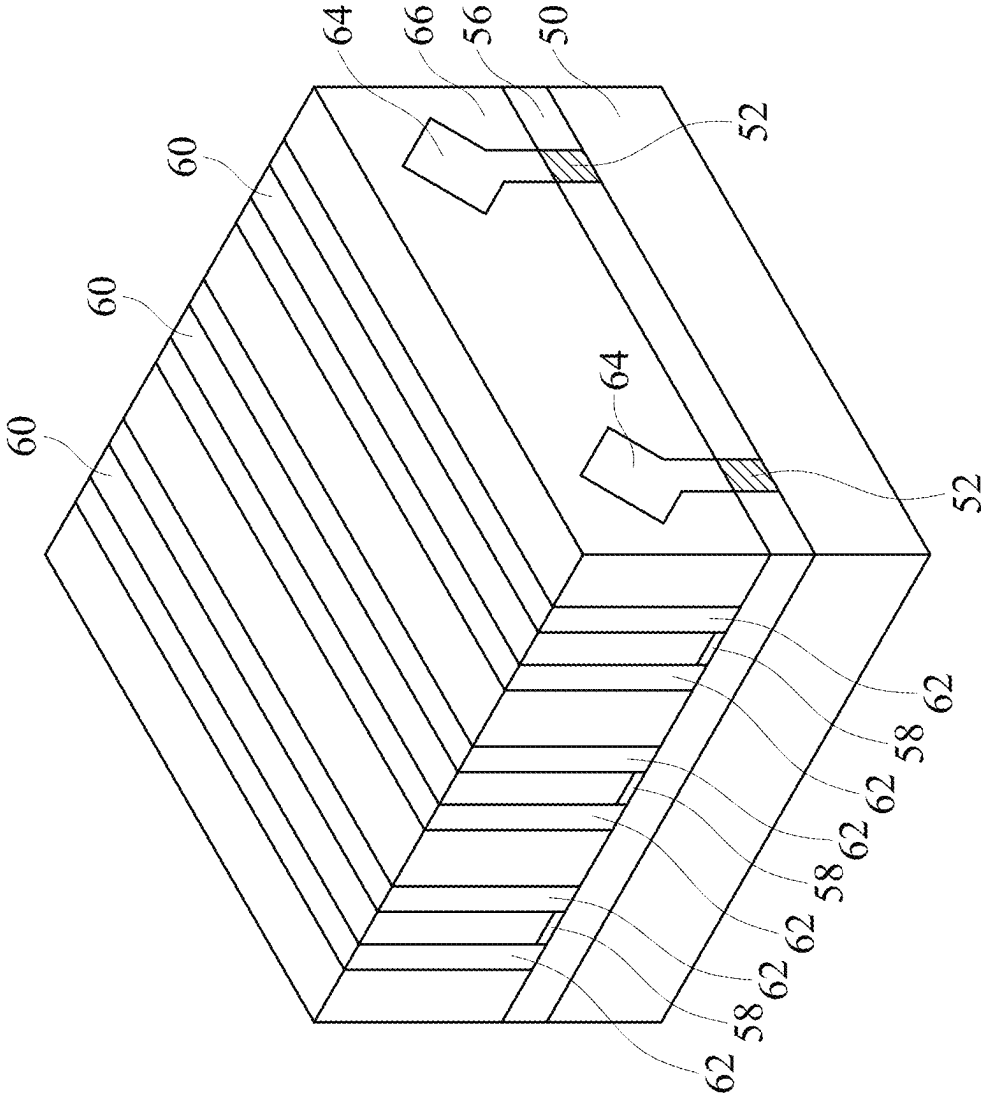


FIG. 3A

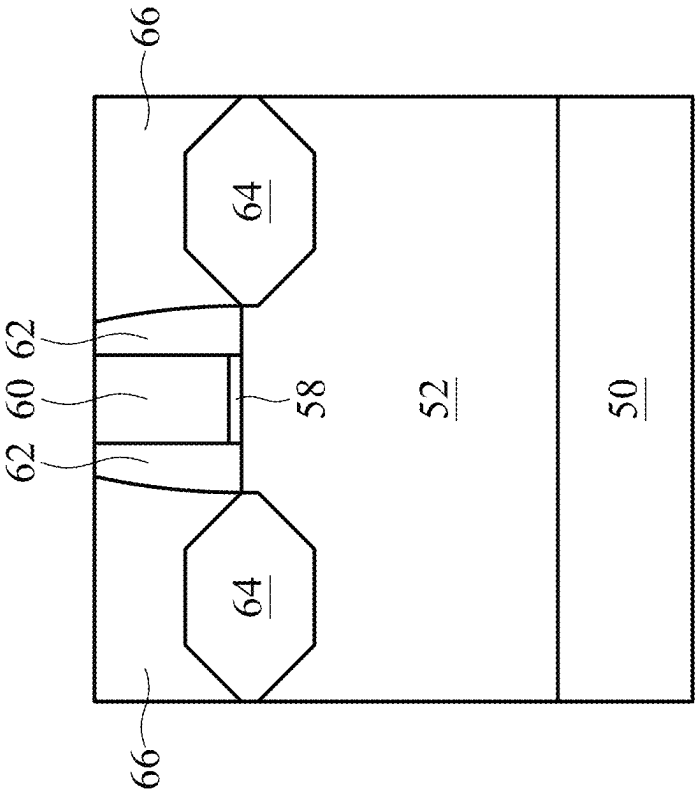


FIG. 3B

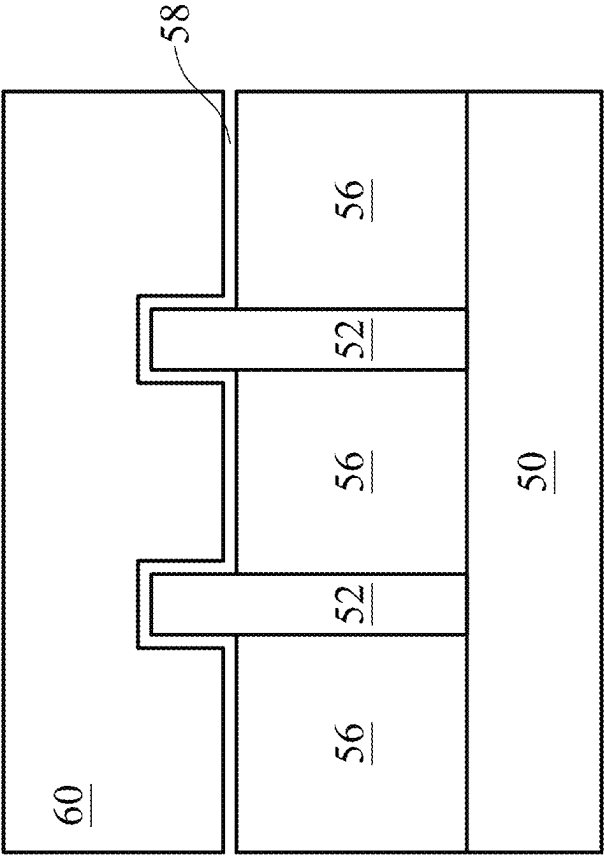


FIG. 3C

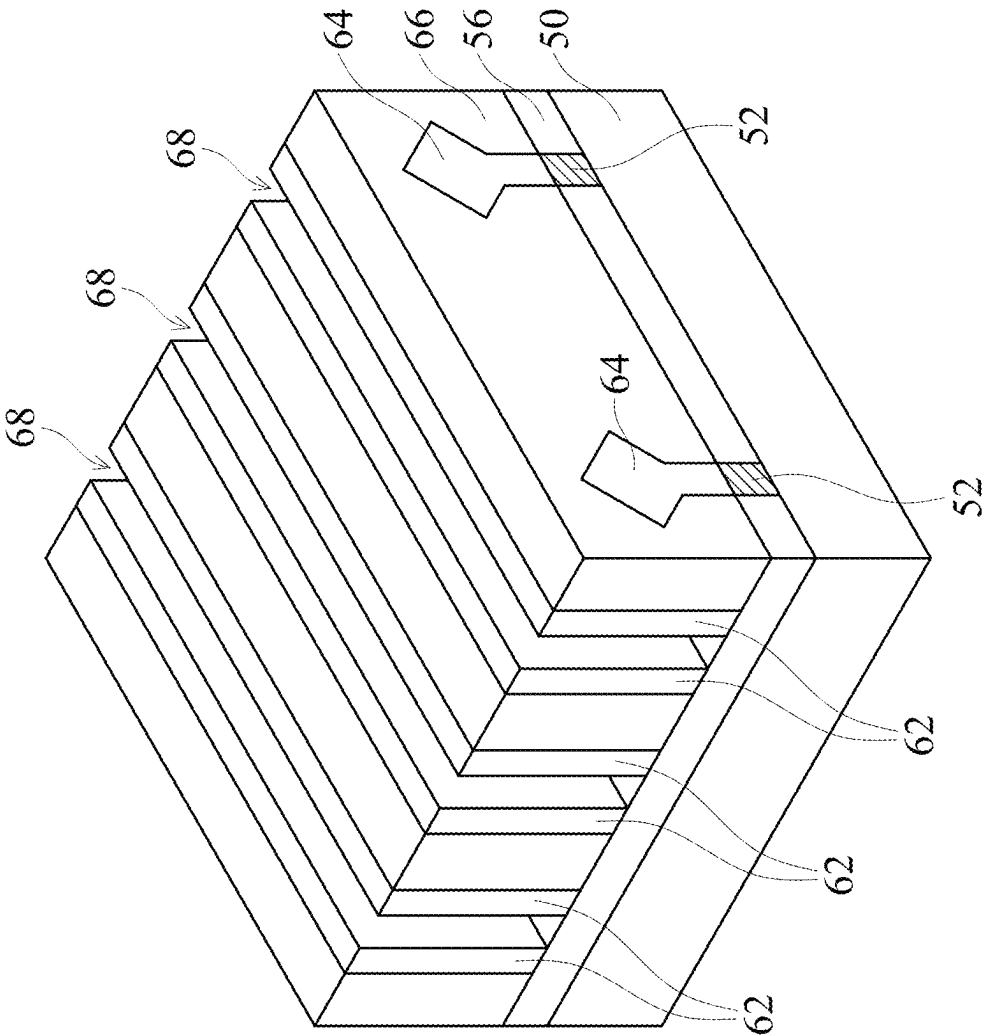


FIG. 4A

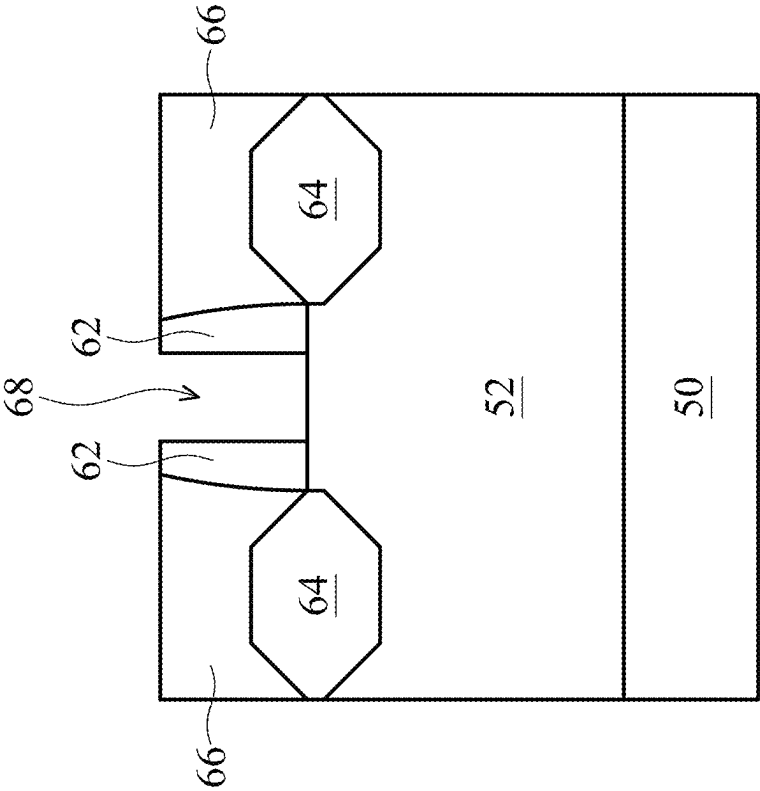


FIG. 4C

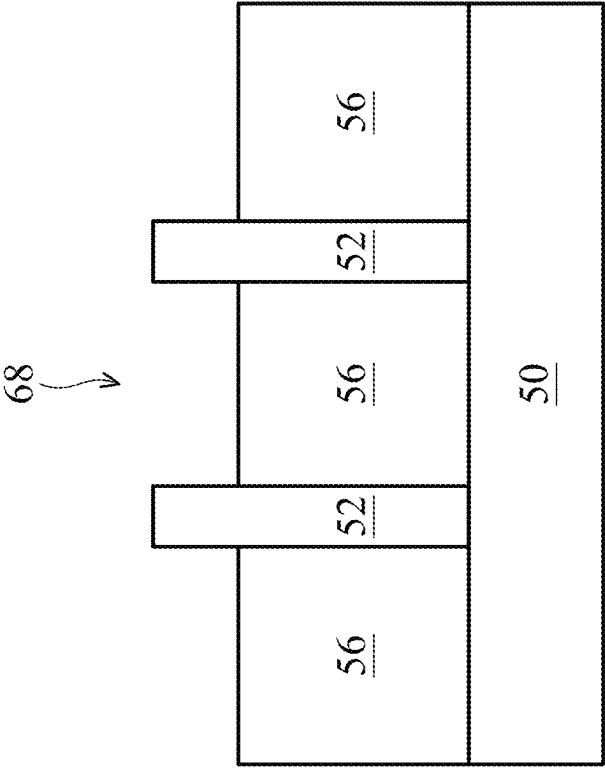


FIG. 4B

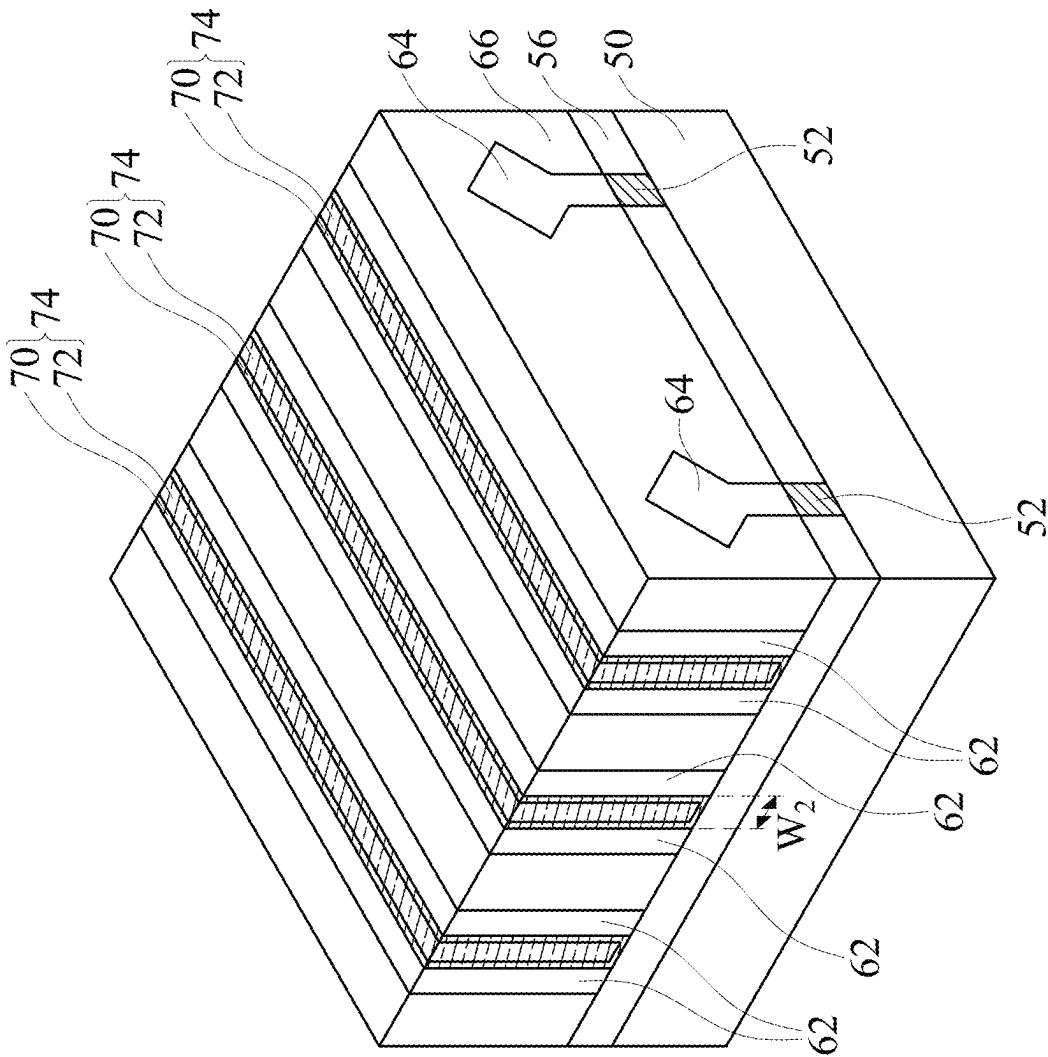


FIG. 5A

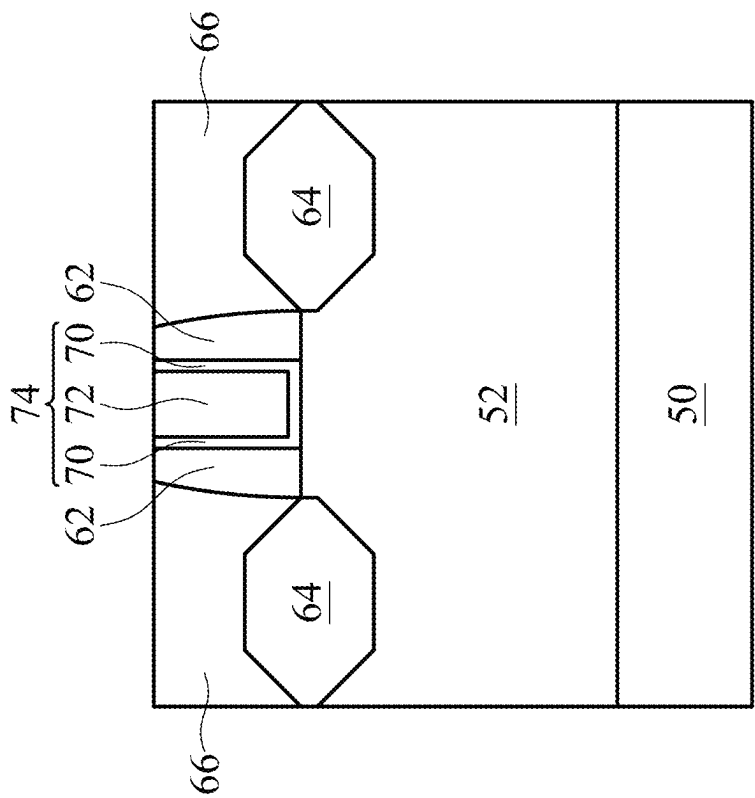


FIG. 5C

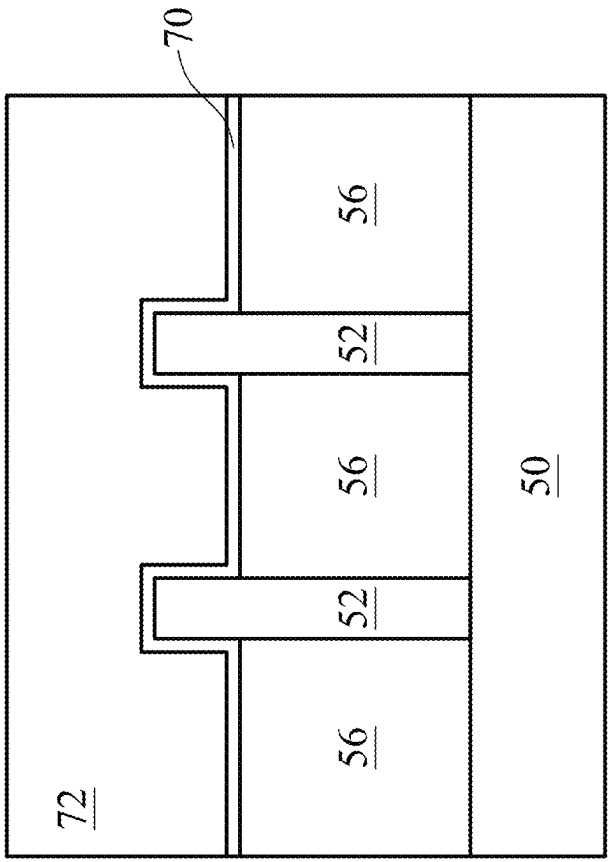


FIG. 5B

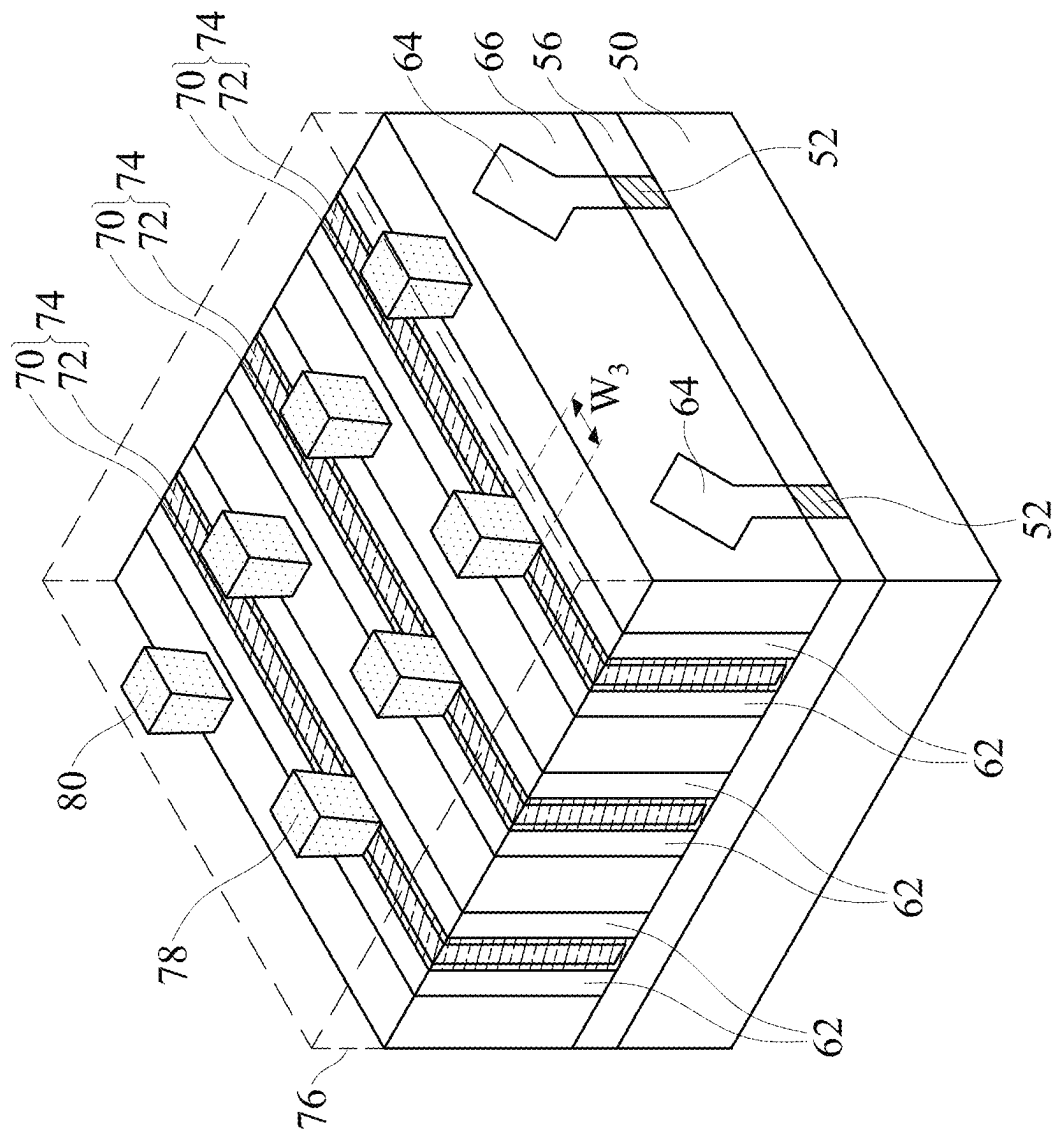


FIG. 6A

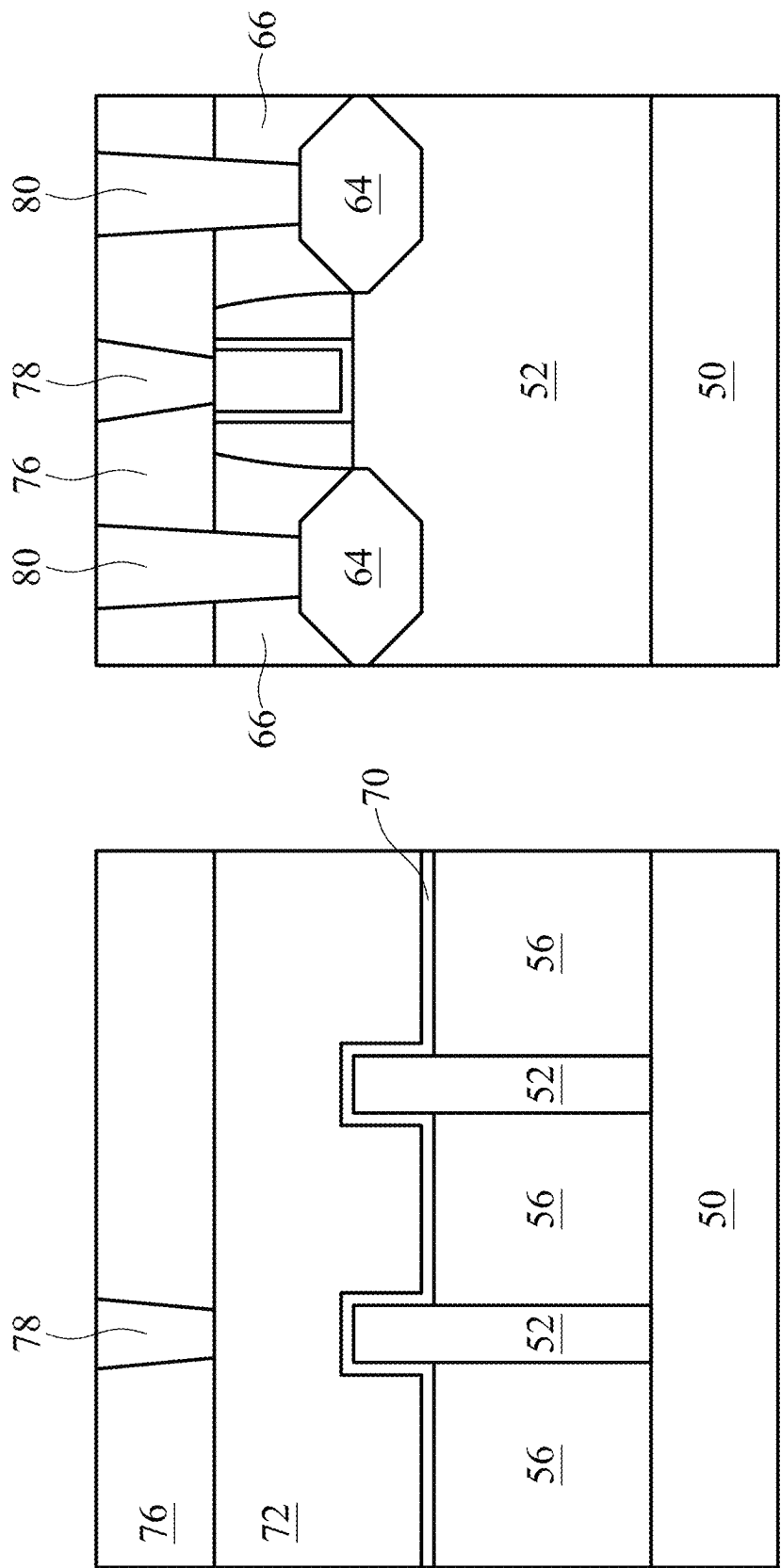


FIG. 6B

FIG. 6C

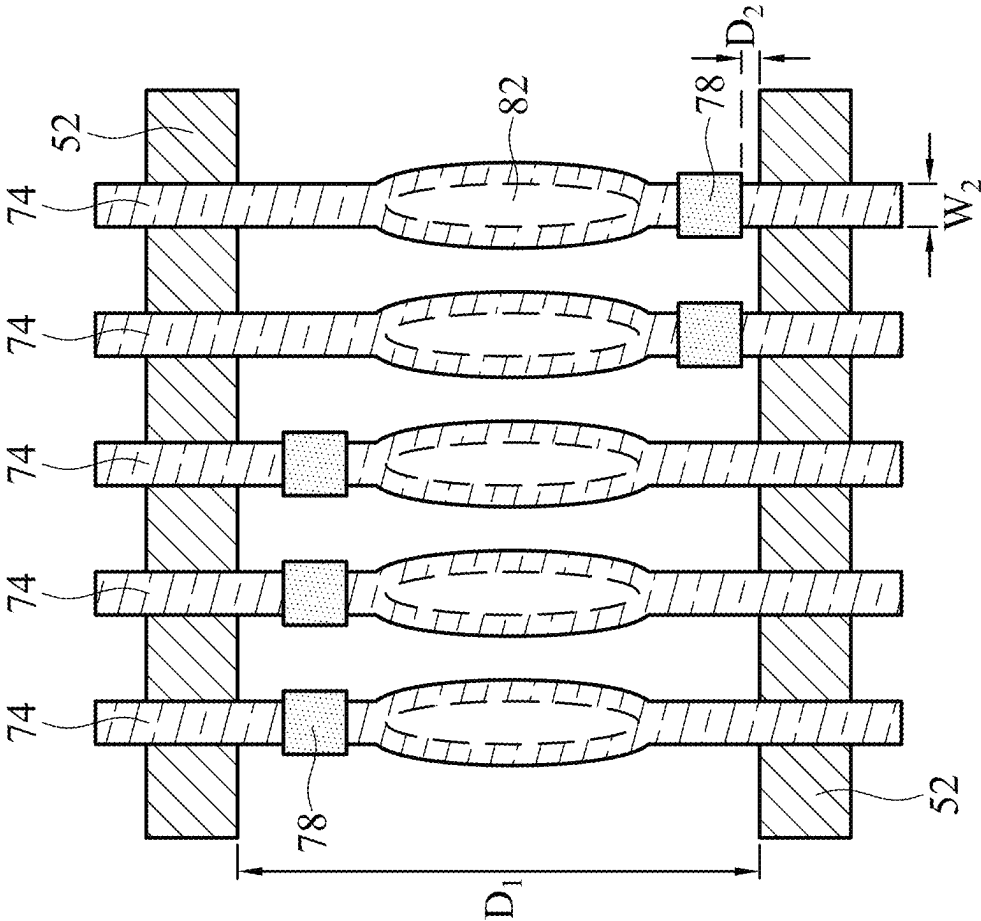


FIG. 7B

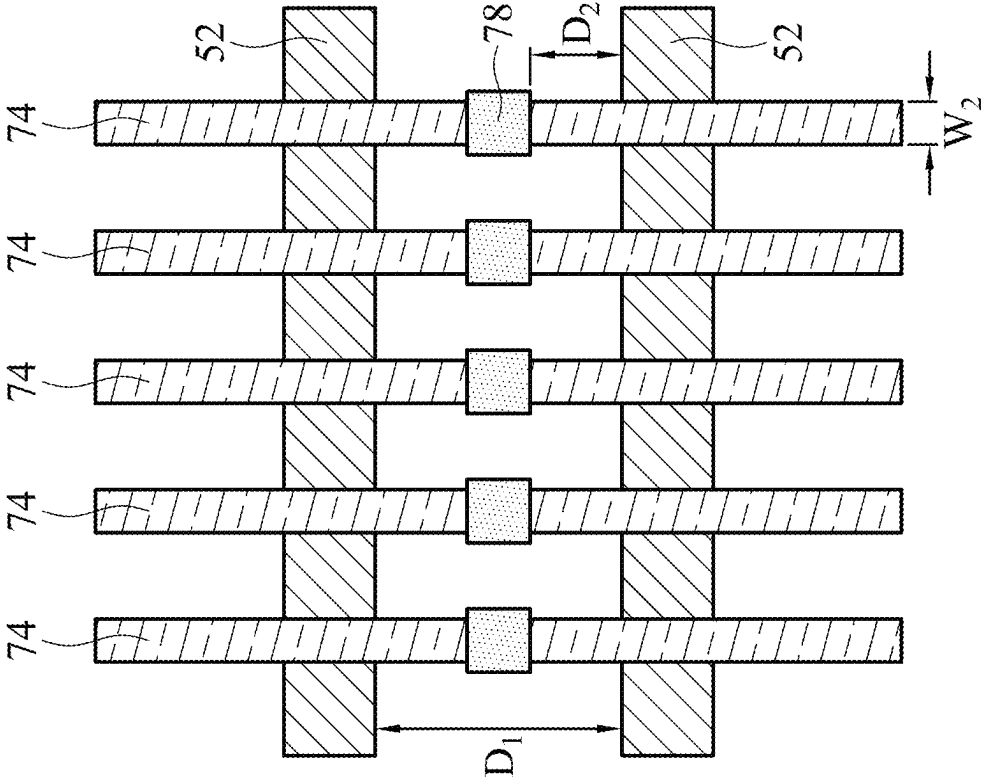


FIG. 7A

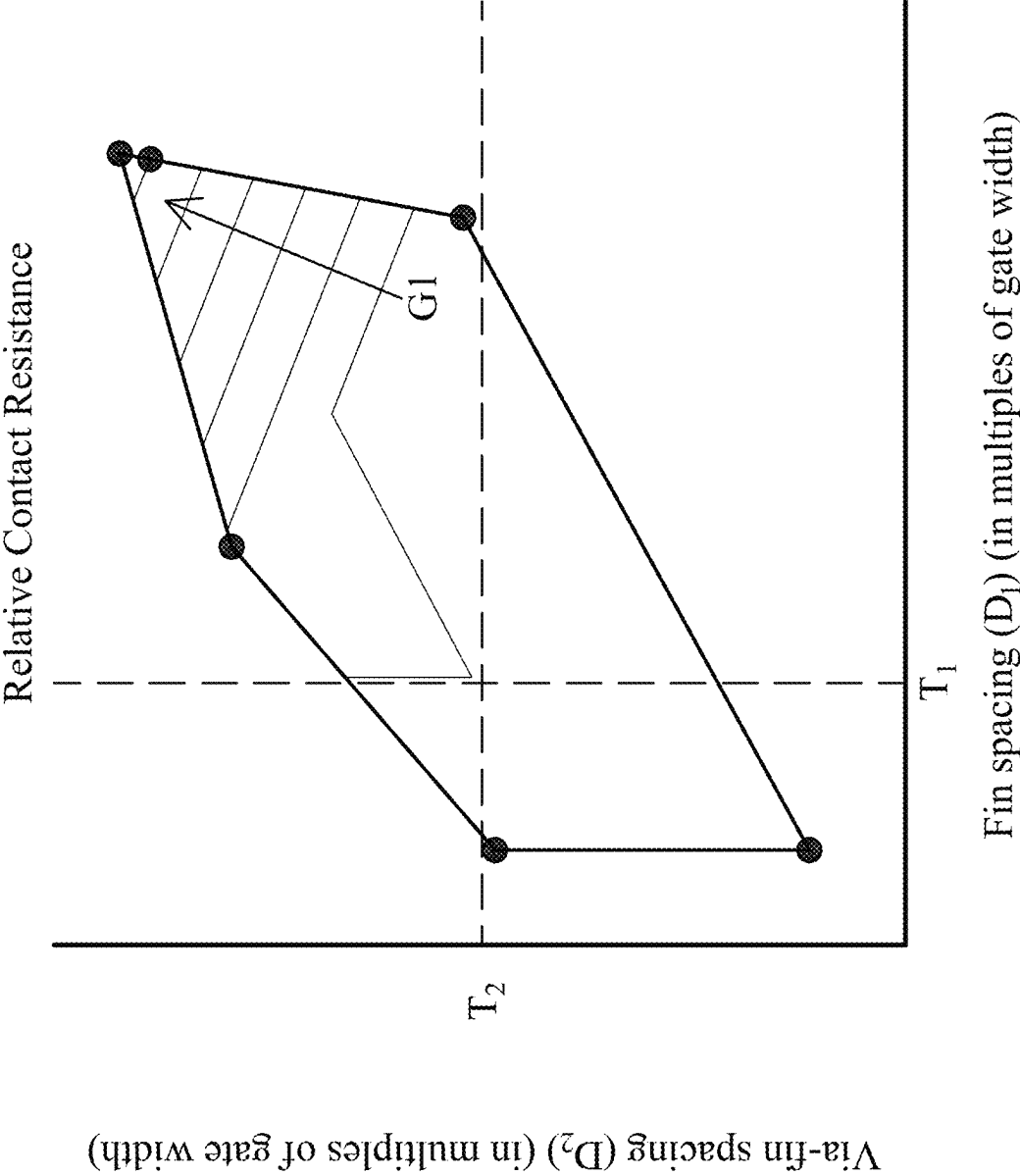


FIG. 8

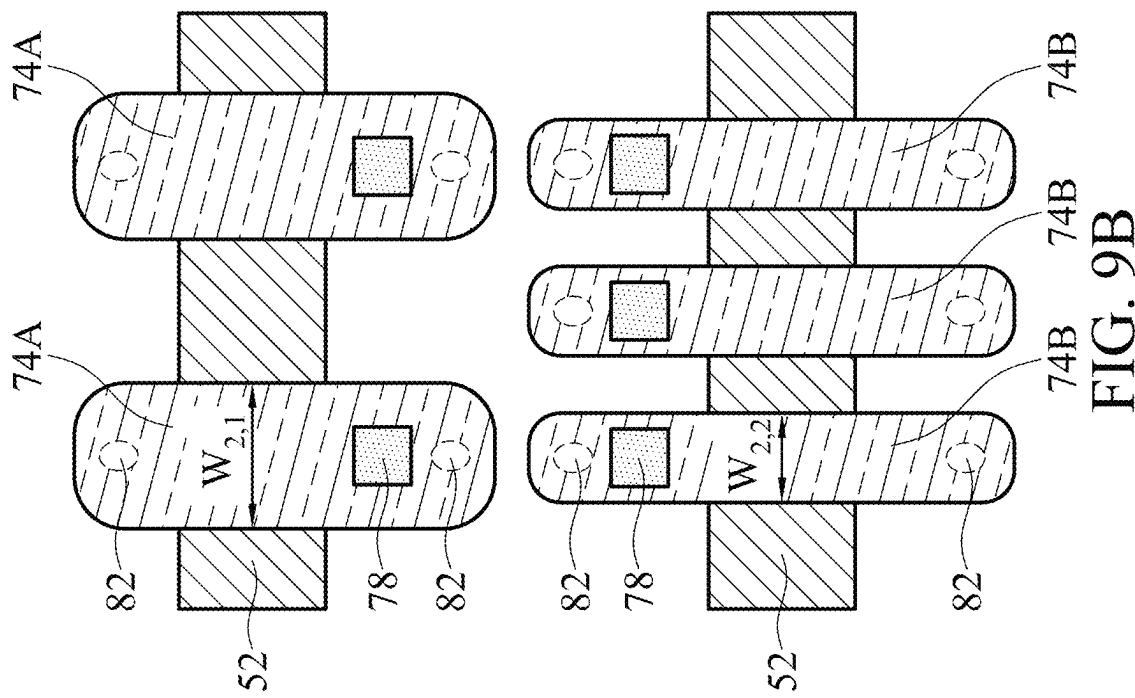


FIG. 9B

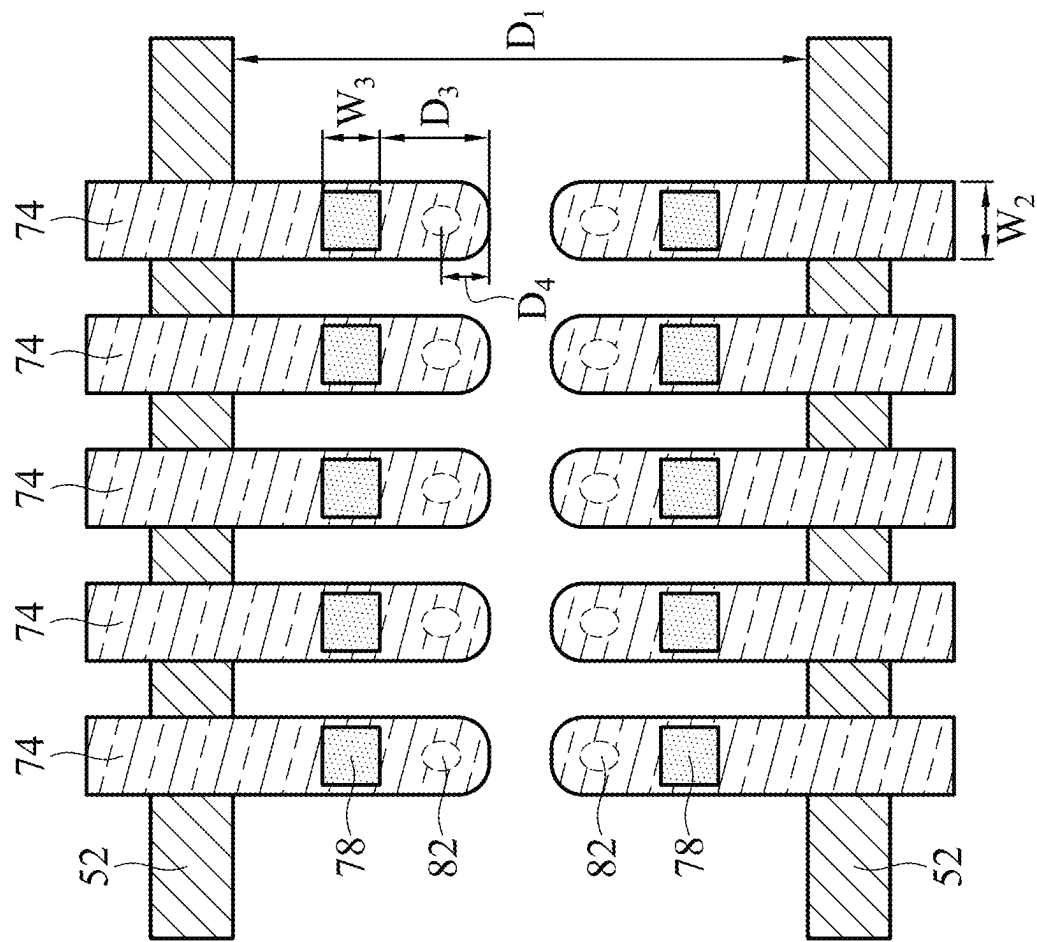


FIG. 9A

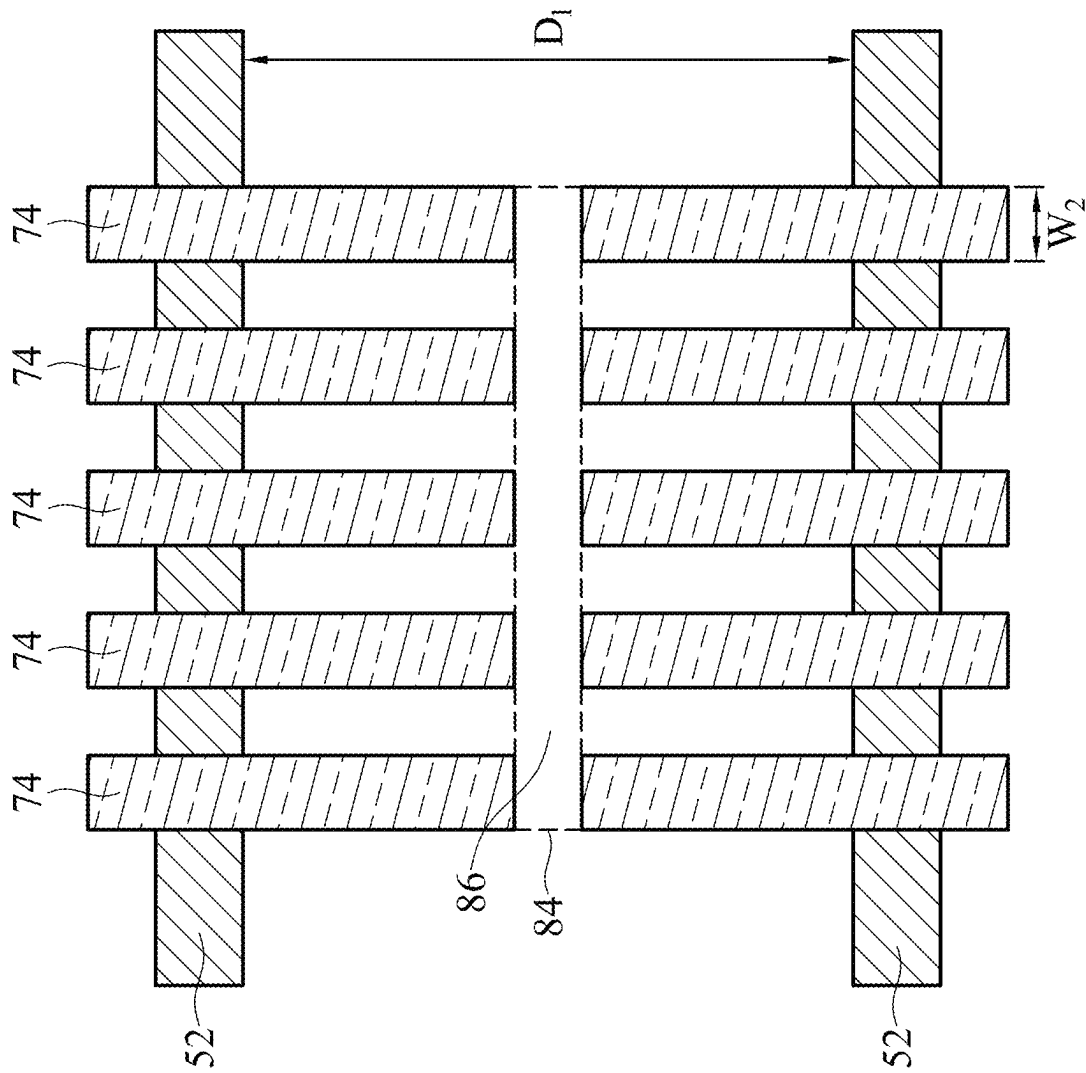


FIG. 10

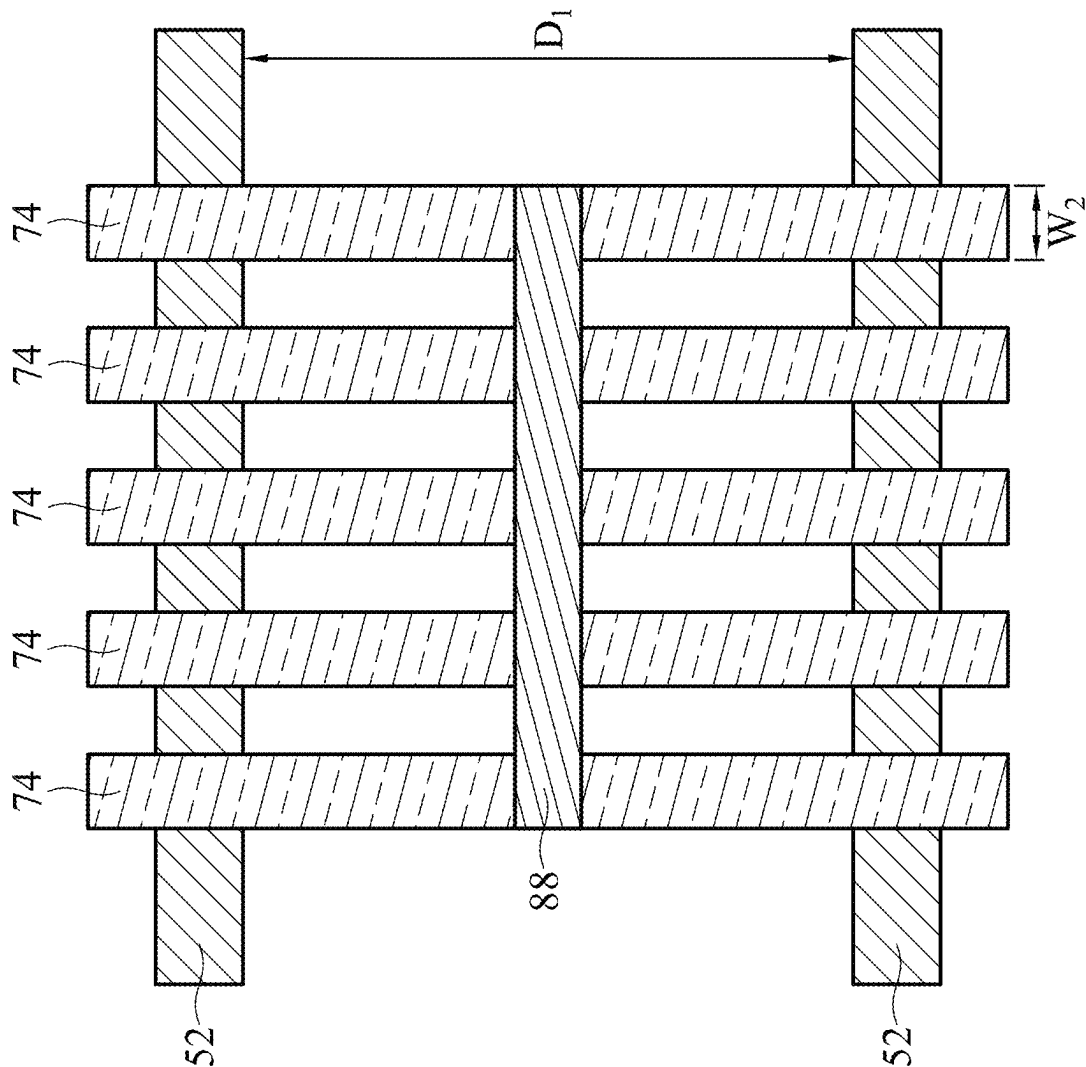


FIG. 11

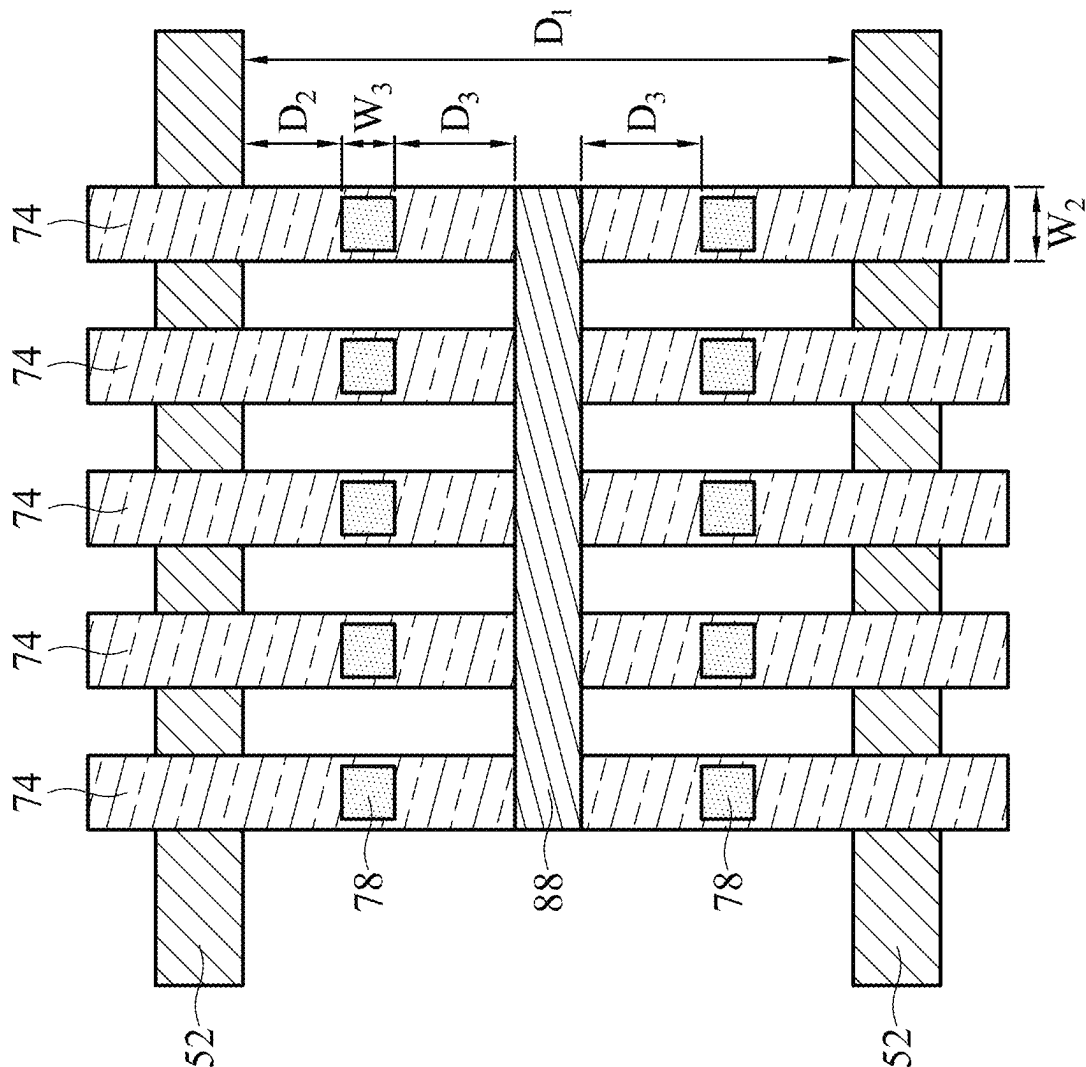


FIG. 12

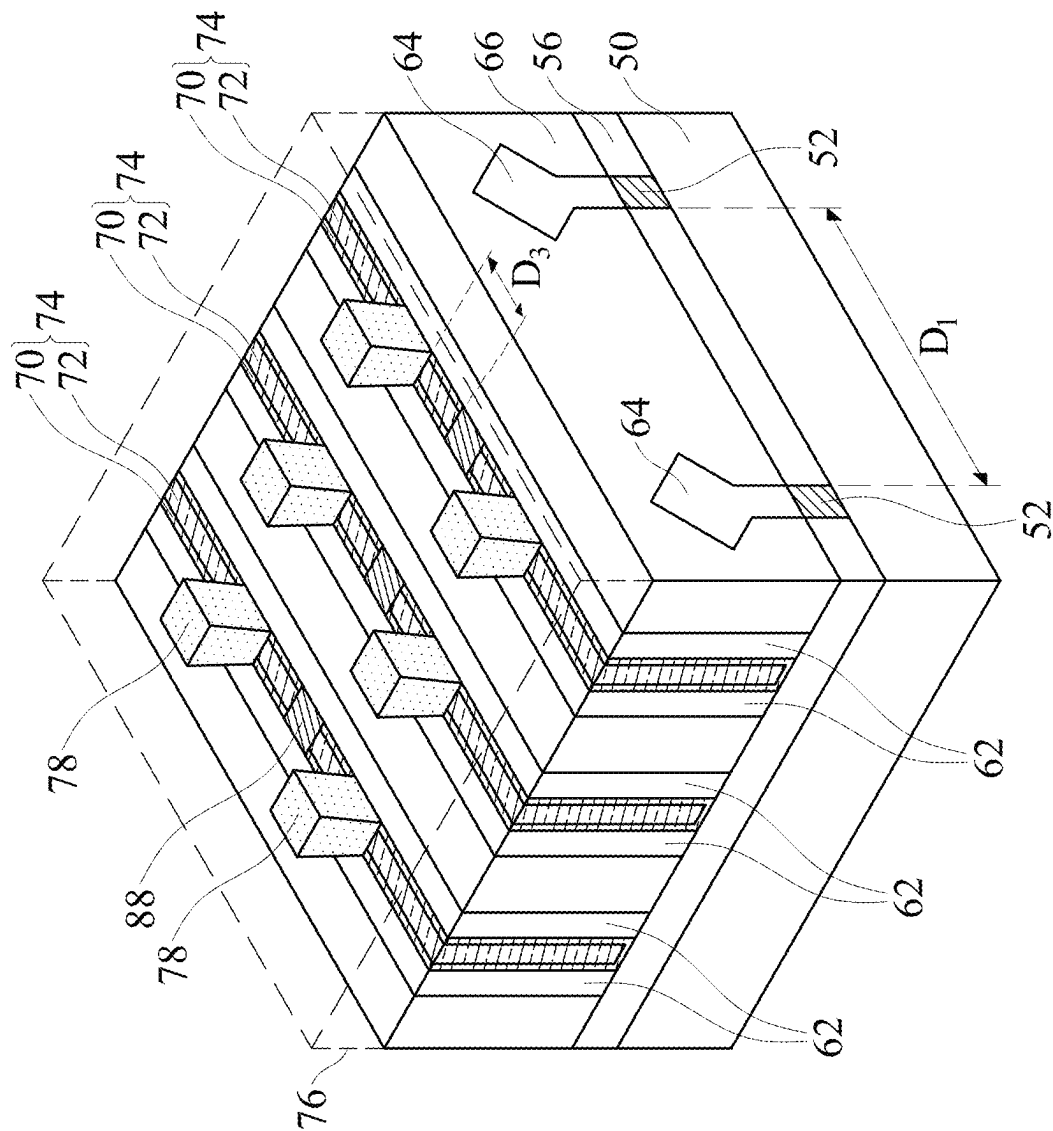


FIG. 13

SEMICONDUCTOR DEVICES

PRIORITY CLAIM AND CROSS-REFERENCE

This application is a continuation of U.S. patent application Ser. No. 18/335,806 filed Jun. 15, 2023, entitled “Semiconductor Device and Method,” which is a continuation of U.S. patent application Ser. No. 17/862,628 filed Jul. 12, 2022, entitled “Semiconductor Device and Method,” now U.S. Pat. No. 11,721,590 issued on Aug. 8, 2023, which is a continuation of U.S. patent application Ser. No. 17/169,809 filed Feb. 8, 2021, entitled “Semiconductor Device and Method,” now U.S. Pat. No. 11,393,724 issued on Jul. 19, 2022, which is a continuation of U.S. patent application Ser. No. 16/567,053 filed Sep. 11, 2019 entitled “Semiconductor Device and Method,” now U.S. Pat. No. 10,957,604 issued on Mar. 23, 2021, which claims the benefit of U.S. Provisional Application No. 62/753,456, filed on Oct. 31, 2018, which applications are hereby incorporated herein by reference.

BACKGROUND

Semiconductor devices are used in a variety of electronic applications, such as, for example, personal computers, cell phones, digital cameras, and other electronic equipment. Semiconductor devices are typically fabricated by sequentially depositing insulating or dielectric layers, conductive layers, and semiconductor layers of material over a semiconductor substrate, and patterning the various material layers using lithography to form circuit components and elements thereon.

The semiconductor industry continues to improve the integration density of various electronic components (e.g., transistors, diodes, resistors, capacitors, etc.) by continual reductions in minimum feature size, which allow more components to be integrated into a given area. However, as the minimum features sizes are reduced, additional problems arise that should be addressed.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIGS. 1A, B, 1C, 2A, 2B, 2C, 2D, 3A, 3B, 3C, 4A, 4B, 4C, 5A, 5B, 5C, 6A, 6B, and 6C illustrate intermediate stages in the manufacturing of FinFETs, in accordance with some embodiments.

FIGS. 7A and 7B illustrate a gate contact layout, in accordance with some embodiments.

FIG. 8 illustrates experimental data for a gate contact layout, in accordance with some embodiments.

FIGS. 9A and 9B illustrate a gate contact layout, in accordance with some other embodiments.

FIGS. 10 through 13 illustrate a gate contact layout, in accordance with some other embodiments.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and

arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

In accordance with some embodiments, metal gate stacks are formed over fins, and gate contacts are formed to the metal gate stacks. In some embodiments, voids form in the metal gate stacks. The spacing between the gate contacts and fins is constrained according to the spacing between adjacent fins. The spacing between the gate contacts and ends of the metal gate stack lines is also constrained. The spacing is constrained to a value that is less than a distance at which voids typically form in the metal gate stack. Formation of the gate contacts on the voids may thus be avoided. Further, in some embodiments, the metal gate stack lines are cut during formation to avoid the formation of voids.

FIGS. 1A through 6C illustrate intermediate stages in the manufacturing of FinFETs, in accordance with some embodiments. FIGS. 1A, 2A, 3A, 4A, 5A, and 6A are three-dimensional views. FIGS. 1B, 2B, 3B, 4B, 5B, and 6B are cross-sectional views shown along a longitudinal axis of the FinFETs, e.g., perpendicular to the direction of current flow between the source/drain regions of the FinFETs, and are shown for a single FinFET. FIGS. 1C, 2C, 3C, 4C, 5C, and 6C are cross-sectional views shown along a latitudinal axis of the FinFETs, e.g., parallel to the direction of current flow between the source/drain regions of the FinFETs, and are shown for a single FinFET. FIG. 2D is a cross-sectional view shown along a longitudinal axis of the FinFETs, through source/drain regions of the FinFETs.

Some embodiments discussed herein are discussed in the context of FinFETs formed using a gate-last process. In other embodiments, a gate-first process may be used. Also, some embodiments contemplate aspects used in planar devices, such as planar FETs.

In FIGS. 1A through 1C, a substrate 50 is provided. The substrate 50 may be a semiconductor substrate, such as a bulk semiconductor, a semiconductor-on-insulator (SOI) substrate, or the like, which may be doped (e.g., with a p-type or an n-type dopant) or undoped. The substrate 50 may be a wafer, such as a silicon wafer. Other substrates, such as a multi-layered or gradient substrate may also be used. One region of the substrate 50 is illustrated, which can be for forming n-type devices (such as NMOS transistors, e.g., n-type FinFETs) or for forming p-type devices (such as PMOS transistors, e.g., p-type FinFETs). The substrate 50

may include multiple, physically separated regions, in which any number of device features (e.g., other active devices, doped regions, isolation structures, etc.) may be formed.

Further, fins **52** are formed extending from the substrate **50**. The fins **52** are semiconductor strips. In the embodiment shown, the fins **52** are an epitaxially grown semiconductor material, different from the material of the substrate **50**. The fins **52** may be formed from silicon, silicon germanium ($\text{Si}_x\text{Ge}_{1-x}$, where x can be in the range of 0 to 1), silicon carbide, pure or substantially pure germanium, a III-V compound semiconductor, a II-VI compound semiconductor, or the like. For example, the available materials for forming III-V compound semiconductor include, but are not limited to, InAs, AlAs, GaAs, InP, GaN, InGaAs, InAlAs, GaSb, AlSb, AlP, GaP, and the like. In the embodiment shown, the fins **52** are formed by epitaxially growing a layer of the semiconductor material on the substrate **50**, and then etching trenches **54** in the semiconductor material, with the fins **52** being formed from portions of the semiconductor material that remain unremoved. The etching may be any acceptable etch process, such as a reactive ion etch (RIE), neutral beam etch (NBE), the like, or a combination thereof. The etch may be anisotropic. In other embodiments, the fins **52** are the same material as the substrate **50**, and are formed by etching trenches in the substrate **50**. As discussed below, the fins **52** are used, to form channel regions of FinFETs. Although only two fins **52** are illustrated, it should be appreciated that any number of fins **52** may be formed.

The fins **52** may be patterned by any suitable method. For example, the fins **52** may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fins.

The fins **52** are formed to a width W_1 at the surface of the substrate **50**. In some embodiments, the width W_1 is in the range of about 6 nm to about 600 nm. Additionally, the fins **52** are spaced apart from each other by a distance D_1 . By spacing the fins **52** in such a fashion, the fins **52** may each form a separate channel region while still being close enough to share a common gate. As discussed further below, the distance D_1 is selected in such a way as to help reduce the contact resistance (R_c) of subsequently formed contacts to the gate of the FinFETs. In some embodiments, the distance D_1 is large, such as in the range of about 22 nm to about 800 nm. In some embodiments, the distance D_1 is small, such as in the range of about 22 nm to about 200 nm.

Further, shallow trench isolation (STI) regions **56** are formed between the fins **52**. The STI regions **56** may be formed by filling the trenches **54** with a dielectric material and recessing the dielectric material in the trenches **54** to form the STI regions **56**. The dielectric material may be an oxide material, a high-density plasma (HDP) oxide, or the like. The dielectric material may be formed, after an optional cleaning and lining of the trenches **54**, using either a chemical vapor deposition (CVD) method, a high density plasma CVD method, or other suitable method of formation as is known in the art.

The trenches **54** may be filled by overfilling the trenches **54** and the substrate **50** with the dielectric material and then removing the excess material outside of the trenches **54** and the fins **52** through a suitable process such as chemical mechanical polishing (CMP), an etch, a combination thereof, or the like. In an embodiment, the removal process removes dielectric material overlying the fins **52**, so that top surfaces of the fins **52** are exposed.

Once the trenches **54** have been filled with the dielectric material, the dielectric material may then be recessed from the top surfaces of the fins **52**. The recessing may be performed to expose at least a portion of the sidewalls of the fins **52** adjacent to the top surfaces of the fins **52**. The dielectric material may be recessed using a wet etch by dipping the top surfaces of the fins **52** into an etchant such as HF, although other etchants, such as H_2 , and other methods, such as a reactive ion etch, a dry etch with etchants such as NH_3/NF_3 , chemical oxide removal, or dry chemical clean may be used. The dielectric material is recessed such that the exposed portions of the fins **52** have a first height H_1 . In some embodiments, the first height H_1 is in the range of about 40 Å to about 100 Å. Additionally, the recessing may also remove any leftover dielectric material located over the fins **52**, exposing the fins **52** for further processing.

In FIGS. 2A through 2C, dummy gate dielectrics **58** and dummy gate electrodes **60** are formed over each of the fins **52**. In some embodiments, a dummy gate dielectric layer is formed by thermal oxidation, chemical vapor deposition, sputtering, or any other methods known and used in the art for forming dielectric layers. A dummy gate electrode layer is then formed on the dummy gate dielectric layer. The dummy gate electrode layer may be formed from a conductive material such as polycrystalline-silicon (polysilicon), poly-crystalline silicon-germanium (poly-SiGe), metallic nitrides, metallic silicides, metallic oxides, metals, or the like, and may be formed by a deposition process such as physical vapor deposition (PVD), chemical vapor deposition (CVD), sputter deposition, or the like. The dummy gate electrode layer and dummy gate dielectric layer are then patterned, such as by acceptable photolithography and etching processes, with remaining portions of the dummy gate dielectric layer and dummy gate electrode layer, respectively, forming the dummy gate dielectrics **58** and dummy gate electrodes **60**.

Further, gate spacers **62** are formed on opposing sides of the dummy gate electrodes **60**, over each of the fins **52**. In some embodiments, the gate spacers **62** are formed by blanket depositing a spacer layer, such as a layer of silicon nitride, silicon oxide, silicon oxynitride, silicon carbide, or the like by a deposition process such as CVD or plasma enhanced CVD (PECVD). The spacer layer is then patterned, such as by one or more etches to remove horizontal portions of the spacer layer, with remaining vertical portions of the spacer layer forming the gate spacers **62**.

Further, epitaxial source/drain regions **64** are formed in the fins **52**, such that each dummy gate electrode **60** is laterally disposed between respective neighboring pairs of the epitaxial source/drain regions **64**. The epitaxial source/drain regions **64** exert stress in what will be the channel regions of the resulting FinFETs, thereby improving performance. The gate spacers **62** separate the epitaxial source/drain regions **64** from the dummy gate electrodes **60** by an appropriate lateral distance so that the epitaxial source/drain regions **64** do not short out subsequently formed gates of the resulting FinFETs. The epitaxial source/drain regions **64** are formed by etching recesses in the fins **52**. Then, the epitaxial source/drain regions **64** in the region are epitaxially grown

in the recesses. The epitaxial source/drain regions **64** may include any acceptable material, such as appropriate for n-type or p-type FinFETs. For example, when n-type FinFETs are formed, the epitaxial source/drain regions **64** may include materials exerting a tensile strain in the channel region of the fins **52**, such as silicon, SiC, SiCP, SiP, or the like. Likewise, when p-type FinFETs are formed, the epitaxial source/drain regions **64** may include materials exerting a compressive strain in the channel region of the fins **52**, such as SiGe, SiGeB, Ge, GeSn, or the like. The epitaxial source/drain regions **64** may have surfaces raised from respective surfaces of the fins **52** and may have facets.

As a result of the epitaxy processes used to form the epitaxial source/drain regions **64**, upper surfaces of the epitaxial source/drain regions **64** have facets which expand laterally outward beyond sidewalls of the fins **52**. In the embodiment shown in FIGS. **2A** through **2C**, adjacent epitaxial source/drain regions **64** remain separated after the epitaxy process is completed. In other embodiments, such as that shown in FIG. **2D**, these facets cause adjacent epitaxial source/drain regions **64** of a same FinFET to merge.

In FIGS. **3A** through **3C**, a first inter-layer dielectric (ILD) **66** is deposited over the substrate **50**. The first ILD **66** may be formed of a dielectric material, and may be deposited by any suitable method, such as CVD, PECVD, or flowable CVD (FCVD). Dielectric materials may include Phospho-Silicate Glass (PSG), Boro-Silicate Glass (BSG), Boron-Doped Phospho-Silicate Glass (BPSG), undoped Silicate Glass (USG), or the like. Other insulation materials formed by any acceptable process may be used. In some embodiments, a contact etch stop layer (CESL) is disposed between the first ILD **66** and the epitaxial source/drain regions **64**, the gate spacers **62**, and the dummy gate electrodes **60**. The CESL may comprise a dielectric material, such as, silicon nitride, silicon oxide, silicon oxynitride, or the like, having a different etch rate than the material of the first ILD **66**. In some embodiments, a planarization process, such as a CMP, is performed to level the top surface of the first ILD **66** with the top surfaces of the dummy gate electrodes **60** and the gate spacers **62**.

In FIGS. **4A** through **4C**, the dummy gate electrodes **60** and dummy gate dielectrics **58** are removed in one or more etching step(s), so that recesses **68** are formed. Each recess **68** exposes a channel region of a respective fin **52**. Each channel region is laterally disposed between neighboring pairs of the epitaxial source/drain regions **64**. During the removal, the dummy gate dielectrics **58** may be used as etch stop layers when the dummy gate electrodes **60** are etched. The dummy gate dielectrics **58** may then be optionally removed after the removal of the dummy gate electrodes **60**.

In FIGS. **5A** through **5C**, gate dielectrics **70** and gate electrodes **72** are formed for replacement gates. The gate dielectrics **70** are conformally deposited in the recesses **68**, such as on the top surfaces and the sidewalls of the fins **52** and on sidewalls of the gate spacers **62**. The gate dielectrics **70** may also be formed on top surfaces of the first ILD **66**. In accordance with some embodiments, the gate dielectrics **70** comprise silicon oxide, silicon nitride, or multilayers thereof. In some embodiments, the gate dielectrics **70** include a high-k dielectric material, and in these embodiments, the gate dielectrics **70** may have a k value greater than about 7.0, and may include a metal oxide or a silicate of Hf, Al, Zr, La, Mg, Ba, Ti, Pb, and combinations thereof. The formation methods of the gate dielectrics **70** may include Molecular-Beam Deposition (MBD), atomic layer deposition, PECVD, and the like. In embodiments where portions of the dummy gate dielectrics **58** remain in the

recesses **68**, the gate dielectrics **70** include a material of the dummy gate dielectrics **58** (e.g., SiO₂).

The gate electrodes **72** are deposited over the gate dielectrics **70**, respectively, and fill the remaining portions of the recesses **68**. The gate electrodes **72** may include a metal-containing material such as TiN, TiO, TaN, TaC, Co, Ru, Al, W, combinations thereof, or multilayers thereof. The gate electrodes **72** may be formed by a deposition process such as atomic layer deposition (ALD). The gate electrodes **72** may include any number of liner layers, work function tuning layers, and fill materials. After the filling of the gate electrodes **72**, a planarization process, such as a CMP, may be performed to remove the excess portions of the gate dielectrics **70** and the material of the gate electrodes **72**, which excess portions are over the top surface of the first ILD **66**. The remaining portions of material of the gate electrodes **72** and the gate dielectrics **70** thus form replacement gates of the resulting FinFETs. The gate electrodes **72** and the gate dielectrics **70** may be collectively referred to as gate stacks **74**. The gate stacks **74** extend along sidewalls of a channel region of the fins **52**.

After formation, the gate stacks **74** have a width W_2 . In some embodiments, the width W_2 is in the range of about 6 nm to about 300 nm. As discussed further below, the distance D_1 between the fins **52** (see FIG. **1A**) is selected according to the width W_2 of the gate stacks **74** that are formed.

In FIGS. **6A** through **6C**, a second ILD **76** is deposited over the first ILD **66**. In some embodiments, the second ILD **76** is a flowable film formed by a flowable CVD method. In some embodiments, the second ILD **76** is formed of a dielectric material such as PSG, BSG, BPSG, USG, or the like, and may be deposited by any suitable method, such as CVD and PECVD.

Further, gate contacts **78** and source/drain contacts **80** are formed through the second ILD **76** and the first ILD **66**. Openings for the source/drain contacts **80** are formed through the first ILD **66** and second ILD **76**, and openings for the gate contacts **78** are formed through the second ILD **76** (and optionally the gate mask, if formed). The gate contacts **78** (or source/drain contacts **80**) may include a liner (such as a diffusion barrier layer, an adhesion layer, or the like) and a conductive material. The openings may be formed using acceptable photolithography and etching techniques. The liner and the conductive material are formed in the openings. The liner may include titanium, titanium nitride, tantalum, tantalum nitride, or the like. The conductive material may be copper, a copper alloy, silver, gold, tungsten, cobalt, aluminum, nickel, or the like. The conductive material may be formed by a deposition process such as CVD. A planarization process, such as a CMP, may be performed to remove excess material from a surface of the second ILD **76**. The remaining liner and conductive material forms the source/drain contacts **80** and gate contacts **78** in the openings. An anneal process may be performed to form a silicide at the interface between the epitaxial source/drain regions **64** and the source/drain contacts **80**. The source/drain contacts **80** are physically and electrically coupled to the epitaxial source/drain regions **64**, and the gate contacts **78** are physically and electrically coupled to the gate electrodes **72**. The source/drain contacts **80** and gate contacts **78** may be formed in different processes, or may be formed in the same process. Although shown as being formed in the same cross-sections, it should be appreciated that each of the source/drain contacts **80** and gate contacts **78** may be formed in different cross-sections, which may avoid shorting of the contacts.

After formation, the gate contacts **78** have a width W_3 . In some embodiments, the width W_3 is in the range of about 10 nm to about 20 nm. As discussed further below, dimensions of the gate stacks **74** are selected according to the width W_3 of the gate contacts **78**.

FIGS. 7A and 7B illustrate a gate contact **78** layout, in accordance with some embodiments. FIGS. 7A and 7B are top-down views, showing some features of the structure of FIG. 6A through 6C. In particular, multiple gate stacks **74** are shown, with each respective gate stack **74** extending over two fins **52**. The gate contacts **78** physically couple the gate stacks **74**, and are disposed a distance D_2 from a closest one of the fins **52**. FIG. 7A shows a scenario where the distance D_1 between the fins **52** is small, e.g., where the distance D_1 is less than a predetermined threshold (discussed further below). As discussed above, a small distance D_1 can be in the range of about 22 nm to about 200 nm. FIG. 7B shows a scenario where the distance D_1 between the fins **52** is large, e.g., where the distance D_1 is greater than the predetermined threshold (discussed further below). As discussed above, a large distance D_1 can be in the range of about 22 nm to about 800 nm.

The width W_2 of the gate stacks **74** is sometimes referred to as the critical dimension of the gate stacks **74**. At smaller critical dimensions, portions of the gate stacks **74** may be deformed. FIG. 7B is a top-down view of deformed gate stacks **74**. The deformed gate stacks **74** each include a void **82**. Portions of the gate stacks **74** deform because the coefficient of thermal expansion (CTE) of the gate stacks **74** is smaller than the CTE of surrounding dielectric materials (e.g., of the first ILD **66** and STI regions **56** (see FIGS. 6A through 6C)). After thermal processing, the surrounding dielectric may shrink more than the fins **52**. The shrinking of the surrounding dielectric pulls apart the gate stacks **74**, causing the voids **82** to be formed. The amount of deformation depends on the volume of the surrounding dielectric. Portions of the gate stacks **74** distal the fins **52** are surrounded by a greater volume of dielectric, and so more deformation of these portions occurs. Portions of the gate stacks **74** proximate the fins **52** are surrounded by a lesser volume of dielectric, and so less (or substantially no) deformation of these portions occurs. When the distance D_1 between fins **52** is small (as in FIG. 7A), substantially no deformation of the gate stacks **74** occurs between the fins **52**. When the distance D_1 between fins **52** is large (as in FIG. 7B), deformation of the gate stacks **74** occurs between the fins **52**.

Because of deformation, the gate stacks **74** have multiple widths. The width W_2 of the gate stacks **74** is the width of the un-deformed portions of the gate stacks **74**, e.g., the widths of the portions over the fins **52**. In other words, the width W_2 is the narrowest width the gate stacks **74**, as measured at the narrowest portions of the gate stacks **74**.

Formation of the voids **82** does not prevent the gate stacks **74** from properly functioning. However, the voids **82** have a high k-value by nature of being air or a vacuum. Forming the gate contacts **78** on the voids **82** results in an increased interface resistance between the gate contacts **78** and gate stacks **74**. The contact resistance (R_c) of the gate contacts **78** is largely determined by the interface resistance between the gate contacts **78** and gate stacks **74**, and that interface resistance may increase with smaller contact areas, e.g., when the gate stacks **74** are small. The contact area may be further reduced when the gate contacts **78** are on the voids **82**, thus increasing the contact resistance of the gate contacts **78**.

In accordance with some embodiments, the gate contacts **78** are formed physically coupling portions of the gate stacks **74** that are substantially free from voids **82**. The contact area of the gate contacts **78** may thus be increased, thereby decreasing the contact resistance of the gate contacts **78**. During a design process, the distance D_2 is determined according to the distance D_1 . In particular, when the distance D_1 is greater than a first predetermined threshold T_1 , the distance D_2 is constrained to be less than a second predetermined threshold T_2 . FIG. 7A shows a scenario where the distance D_1 is less than the first predetermined threshold T_1 , and so the distance D_2 is unconstrained. FIG. 7B shows a scenario where the distance D_1 is greater than the first predetermined threshold T_1 , and so the distance D_2 is constrained. The first predetermined threshold T_1 and second predetermined threshold T_2 are defined as multiples of the width W_2 of the gate stacks **74**. In some embodiments, the first predetermined threshold T_1 is in a range of about 35 to about 40 times the width W_2 of the gate stacks **74**, and the second predetermined threshold T_2 is in a range of about 10 to about 13 times the width W_2 of the gate stacks **74**. Further, a ratio of the first predetermined threshold T_1 to the second predetermined threshold T_2 can be in the range of about 3 to about 5.

FIG. 8 is a heat map showing experimental data for the gate contact **78** layout of FIGS. 7A and 7B. In FIG. 8, the X-axis plots spacing between adjacent fins **52** (e.g., the distance D_1), the Y-axis plots spacing between a gate contact **78** and an adjacent fin **52** (e.g., the distance D_2), and values on the heat map plot the relative contact resistance. For example, the X-axis can be in a range of about 15 to about 75, and the Y-axis can be in a range of about 5 to about 20. As shown, when the distance D_1 was less than the first predetermined threshold T_1 , all values for the distance D_2 resulted in a low contact resistance. However, when the distance D_1 was greater than the first predetermined threshold T_1 , values for the distance D_2 that were greater than the second predetermined threshold T_2 resulted in a rapidly-increasing relative contact resistance along gradient G_1 . For example, in embodiments where the width W_2 is about 6 nm, the distance D_2 is constrained to being less than about 70 nm when the distance D_1 is greater than or equal to about 216 nm, and the distance D_2 is unconstrained when the distance D_1 is less than about 216 nm. Constraining the distances according to the experimental results shown in FIG. 8 allows the gate contacts **78** to be placed without substantially increasing the contact resistance.

The embodiments described in FIGS. 7A and 7B may be combined on a same substrate. For example, in a first region of the substrate, the distance D_1 between fins **52** may be constrained to less than the first predetermined threshold T_1 . Such a constraint may prevent or reduce the formation of voids **82** in the first region. Likewise, in a second region of the substrate, the distance D_1 between fins **52** may be greater than the first predetermined threshold T_1 , and the distance D_2 between the fins **52** and gate contacts **78** is constrained to less than the second predetermined threshold T_2 . Such a constraint may help avoid forming the gate contacts **78** on voids **82** in the second region.

As used herein, "constraining" the widths W_1 and W_2 refers to imposing constraints during a design process for the fins **52** and gate contacts **78**. After the design process is completed, the design may be stored in, e.g., a cell library. The cell library may then be used to manufacture a corresponding FinFET.

FIG. 9A illustrates a gate contact **78** layout, in accordance with some other embodiments. FIG. 9A is a top-down view,

showing some features of the structure of FIGS. 6A through 6C. In particular, two of the fins 52 are shown, with each respective fin 52 being adjacent the end of the metal lines that form the gate stacks 74. The gate contacts 78 physically couple the gate stacks 74, and are disposed a distance D_3 from the ends of the metal lines.

When forming the gate stacks 74, voids 82 may also be formed at the ends of the metal lines. As noted above, the dummy gate dielectrics 58 and dummy gate electrodes 60 are replaced by forming recesses 68 exposing channel regions of the fins 52, and forming the gate stacks 74 in the recesses 68. The etching step(s) for forming the recesses 68 results in the recesses 68 having rounded ends. As noted above, the gate stacks 74 are formed by depositing a conductive material in the recesses 68, such as with an ALD process. The rounded ends of the recesses 68 are narrower than the main portions of the recesses 68, and the ALD process may have imperfect gap-filling properties. As such, voids 82 may be formed at ends of the gate stacks 74, proximate the rounded ends. The voids 82 are disposed a distance D_4 from the ends of the metal lines. In some embodiments, the distance D_4 is in the range of about 50 nm to about 90 nm.

In accordance with some embodiments, the gate contacts 78 are formed physically coupling portions of the gate stacks 74 that are substantially free from voids 82. The contact resistance of the gate contacts 78 may thus be decreased. During a design process, the distance D_3 is determined according to the width W_2 of the gate stacks 74 and the width W_3 of the gate contacts 78. In particular, the distance D_3 is greater than the distance D_4 , and at least the sum of half of the width W_2 and half of the width W_3 , e.g., D_3 is constrained according to $D_3 > D_4$ and $D_3 > 0.5 * W_2 + 0.5 * W_3$. In some embodiments, the distance D_3 is in the range of about 91 nm to about 500 nm. As such, when the gate contacts 78 are formed, the voids 82 are laterally disposed between the gate contacts 78 and the ends of the metal lines. The distance D_3 is large enough to help avoid formation of the gate contacts 78 on the voids 82.

The distance D_1 between fins 52 at the ends of the metal lines may be larger, to accommodate the larger distance D_3 . In particular, the distance D_1 in the embodiment of FIG. 9A is larger than the distance D_1 in the embodiments of FIGS. 7A and 7B. In some embodiments, the distance D_1 between fins 52 at the ends of the metal lines is in the range of about 442 nm to about 1000 nm.

FIG. 9B illustrates a gate contact 78 layout, in accordance with some other embodiments. FIG. 9B is a top-down view, showing some features of the structure of FIGS. 6A through 6C. The embodiment of FIG. 9B is similar to the embodiment of FIG. 9A, but includes gate stacks 74 of multiple widths. For example, a first set of gate stacks 74A may have a width $W_{2,1}$, and a second set of gate stacks 74B may have a width $W_{2,2}$, with the width $W_{2,1}$ being greater than the width $W_{2,2}$. In some embodiments, the width $W_{2,1}$ is in the range of about 67 nm to about 151 nm, and the width $W_{2,2}$ is in the range of about 67 nm to about 151 nm.

FIGS. 10 through 12 are top-down views of intermediate stages in a process for forming the gate contacts 78, in accordance with some embodiments. FIG. 13 is a three-dimensional view of the resulting structure. Multiple gate stacks 74 are shown, with each respective gate stack 74 initially being formed as a continuous metal line extending over two fins 52. The gate stacks 74 may be formed to originally span the fins 52 of multiple transistors (where each transistor may include one or a plurality of fins 52). The distance D_1 between fins 52 is large, to accommodate a line

cut that will be subsequently performed. In particular, the distance D_1 in the embodiment of FIGS. 10 through 13 is larger than the distance D_1 in the embodiments of FIGS. 7A and 7B. In some embodiments, the distance D_1 between fins 52 is in the range of about 350 nm to about 1000 nm.

In FIG. 10, each respective gate stack 74 is cut in a region 84 to form an opening 86. The opening 86 can also be formed in dielectric layers around the gate stacks 74, such as the first ILD 66. After the cut, the continuous metal lines are broken into smaller metal lines. The cut may be performed by acceptable photolithography and etching processes. For example, a photoresist may be formed, and patterned to expose the region 84. One or more etching processes may then be performed on the gate stacks 74, using the patterned photoresist as an etching mask. The photoresist may then be removed, such as by an ashing process. After being cut, each gate stack 74 may span the fins 52 of a single transistor (where the transistor may include one or a plurality of fins 52). By forming a continuous metal line and cutting it, the formation of voids at the ends of the metal lines may be avoided. The ends of each gate stack 74 are thus free from voids.

In FIG. 11, the opening 86 is filled with a dielectric material 88. The dielectric material 88 may be silicon nitride, silicon oxynitride, silicon oxide, silicon carbide, or the like, and may be formed by blanket depositing a dielectric layer and planarizing the dielectric layer to remove portions of the dielectric material 88 outside of the opening 86. Remaining portions of the dielectric material 88 isolate the cut metal lines from one another.

In FIG. 12, the gate contacts 78 are formed, contacting the cut gate stacks 74. The gate contacts 78 physically couple the cut gate stacks 74, and are disposed a same distance D_3 from the ends of the metal lines (e.g., from the dielectric material 88). Because the formation of voids has been avoided, the distance D_3 in the embodiment of FIGS. 10 through 13 may be smaller than the distance D_3 in the embodiments of FIGS. 9A and 9B. In some embodiments, the distance D_3 is in the range of about 91 nm to about 500 nm. The gate contacts 78 may be formed at the ends of the cut gate stacks 74 without the need to avoid contacting voids.

Embodiments may achieve advantages. By constraining the distance D_2 between the fins 52 and gate contacts 78 (see FIG. 7B) for fins 52 spaced by a large distance D_1 , formation of the gate contacts 78 on voids 82 may be avoided. By increasing the distance D_3 between the gate contacts 78 and the ends of the gate stack metal lines, formation of the gate contacts 78 on voids 82 may be further avoided. The contact area of the gate contacts 78 may thus be increased, thereby decreasing the contact resistance of the gate contacts 78. Finally, by initially forming longer gate stack metal lines and cutting the lines, the formation of voids may be avoided, and the gate contacts 78 may be formed at the ends of the cut gate stacks 74 without the need to avoid contacting voids.

In an embodiment, a method includes: forming a first fin extending from a substrate; forming a second fin extending from the substrate, the second fin being spaced apart from the first fin by a first distance; forming a metal gate stack over the first fin and the second fin; depositing a first inter-layer dielectric over the metal gate stack; and forming a gate contact extending through the first inter-layer dielectric to physically contact the metal gate stack, the gate contact being laterally disposed between the first fin and the second fin, the gate contact being spaced apart from the first fin by a second distance, where the second distance is less

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than a second predetermined threshold when the first distance is greater than or equal to a first predetermined threshold.

In some embodiments of the method, the metal gate stack has multiple widths, and a portion of the metal gate stack with a narrowest width of the multiple widths is disposed over the first fin. In some embodiments of the method, the first predetermined threshold is in a range of 35 to 40 times the narrowest width of the metal gate stack. In some embodiments of the method, the second predetermined threshold is in a range of 10 to 13 times the narrowest width of the metal gate stack. In some embodiments of the method, the narrowest width of the metal gate stack is in a range of 6 nm to 300 nm. In some embodiments, the method further includes: depositing a second inter-layer dielectric around the metal gate stack, where after the second inter-layer dielectric is formed, a first portion of the metal gate stack includes a void. In some embodiments of the method, the gate contact physically contacts a second portion of the metal gate stack, the first portion of the metal gate stack being distal the first fin, the second portion of the metal gate stack being proximate the first fin. In some embodiments of the method, the first portion of the metal gate stack and the second portion of the metal gate stack are laterally disposed between the first fin and the second fin.

In an embodiment, a method includes: forming a first fin extending from a substrate; forming a dummy gate stack over the first fin; depositing a first inter-layer dielectric along the dummy gate stack; replacing the dummy gate stack with a metal gate stack, an end of the metal gate stack being rounded, the metal gate stack including a void, the void being disposed a first distance from the rounded end of the metal gate stack; forming a second inter-layer dielectric over the metal gate stack and the first inter-layer dielectric; and forming a gate contact extending through the second inter-layer dielectric to physically contact a first portion of the metal gate stack, the first portion of the metal gate stack being disposed a second distance from the rounded end of the metal gate stack, the second distance being greater than the first distance.

In some embodiments of the method, the first distance is in a range of 50 nm to 90 nm, and the second distance is in a range of 91 nm to 500 nm. In some embodiments of the method, the first fin has a first width, the gate contact has a second width, and the second distance is greater than a sum of half of the first width and half of the second width. In some embodiments of the method, the first fin is adjacent the rounded end of the metal gate stack. In some embodiments of the method, replacing the dummy gate stack with the metal gate stack includes: etching the dummy gate stack to form a recess having a rounded end; and filling the recess with a metal, the void being formed during the filling.

In an embodiment, a method includes: forming a first fin extending from a substrate; forming a second fin extending from the substrate; forming a metal line over the first fin and the second fin; cutting the metal line along a first region to separate the metal line into a first metal gate stack and a second metal gate stack, the first metal gate stack being over the first fin, the second metal gate stack being over the second fin; depositing a first inter-layer dielectric over the first metal gate stack and the second metal gate stack; forming a first gate contact extending through the first inter-layer dielectric to physically contact the first metal gate stack, the first gate contact being disposed a first distance from the first region; and forming a second gate contact extending through the first inter-layer dielectric to physically

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contact the second metal gate stack, the second gate contact being disposed the first distance from the first region.

In some embodiments of the method, cutting the metal line includes: etching the metal line in the first region to form a first opening through the metal line; and forming a dielectric material in the first opening. In some embodiments of the method, the first inter-layer dielectric is further deposited over the dielectric material. In some embodiments, the method further includes: forming gate spacers adjacent the metal line, the dielectric material extending between the gate spacers. In some embodiments, the method further includes: forming a second inter-layer dielectric around the gate spacers. In some embodiments of the method, the first fin and the first metal gate stack are part of a first transistor, and where the second fin and the second metal gate stack are part of a second transistor. In some embodiments of the method, ends of the first metal gate stack and the second metal gate stack are free from voids.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A device comprising:

- a first gate structure;
- a second gate structure, the first gate structure and the second gate structure being longitudinally aligned, a first end of the first gate structure facing a second end of the second gate structure;
- an inter-layer dielectric over the first gate structure and the second gate structure;
- a first gate contact extending through the inter-layer dielectric, the first gate contact physically contacting the first gate structure, the first gate contact being disposed a first distance from the first end of the first gate structure; and
- a second gate contact extending through the inter-layer dielectric, the second gate contact physically contacting the second gate structure, the second gate contact being disposed a second distance from the second end of the second gate structure, the first distance and the second distance being substantially equal.

2. The device of claim 1, further comprising:

- a dielectric material physically contacting the first end of the first gate structure and the second end of the second gate structure.

3. The device of claim 2, wherein:

the first gate structure comprises:

- a first gate dielectric in contact with the dielectric material; and
- a first gate electrode in contact with the dielectric material; and

the second gate structure comprises:

- a second gate dielectric in contact with the dielectric material; and
- a second gate electrode in contact with the dielectric material.

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4. The device of claim 1, further comprising:
a first channel region under the first gate structure; and
a second channel region under the second gate structure.
5. The device of claim 4, further comprising:
an isolation region between the first channel region and
the second channel region, the first gate structure
disposed on a top surface of the isolation region, the
second gate structure disposed on the top surface of the
isolation region.
6. The device of claim 4, further comprising:
a first source/drain region adjacent the first channel
region; and
a second source/drain region adjacent the second channel
region, the first source/drain region separated from the
second source/drain region.
7. A device comprising:
a first gate structure;
a second gate structure;
a dielectric material between the first gate structure and
the second gate structure, a top surface of the dielectric
material being planar with a top surface of the first gate
structure and a top surface of the second gate structure;
an inter-layer dielectric over the top surface of the dielec-
tric material, the top surface of the first gate structure,
and the top surface of the second gate structure;
a first gate contact extending through the inter-layer
dielectric to physically contact the first gate structure,
the first gate contact being disposed a first distance
from the dielectric material, the first distance being in
a range of about 91 nm to about 500 nm; and
a second gate contact extending through the inter-layer
dielectric to physically contact the second gate struc-
ture, the second gate contact being disposed a second
distance from the dielectric material, the second dis-
tance being in a range of about 91 nm to about 500 nm.
8. The device of claim 7, further comprising:
a first channel region, the first gate structure extending
along multiple sides of the first channel region; and
a second channel region, the second gate structure extend-
ing along multiple sides of the second channel region.
9. The device of claim 8, wherein the first gate contact and
the second gate contact are each disposed between the first
channel region and the second channel region.
10. The device of claim 8, wherein the first channel region
and the second channel region are parts of different transis-
tors.
11. The device of claim 7, wherein ends of the first gate
structure and the second gate structure are free from voids.
12. The device of claim 7, wherein the first gate structure
has a first width, the second gate structure has a second
width, the first gate contact has a third width, the second gate
contact has a fourth width, the first distance is greater than
a sum of half of the first width and half of the third width,
and the second distance is greater than a sum of half of the
second width and half of the fourth width.

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13. The device of claim 12, wherein the third width and
the fourth width are substantially equal.
14. The device of claim 7, wherein the first distance and
the second distance are substantially equal.
15. A device comprising:
a first gate structure;
a second gate structure, a first end of the first gate
structure facing a second end of the second gate struc-
ture;
an inter-layer dielectric over the first gate structure and the
second gate structure;
a first gate contact extending through the inter-layer
dielectric, the first gate contact physically contacting
the first gate structure at a first contact point, the first
gate contact being disposed a first distance from the
first end of the first gate structure; and
a second gate contact extending through the inter-layer
dielectric, the second gate contact physically contacting
the second gate structure at a second contact point, the
second gate contact being disposed a second distance
from the second end of the second gate structure,
wherein a line extending through the first contact point
and the second contact point is parallel to a first
longitudinal axis of the first gate structure and to a
second longitudinal axis of the second gate structure,
wherein half a distance between the first contact point
and the second contact point is larger than the first
distance and is larger than the second distance.
16. The device of claim 15, further comprising:
a dielectric material physically contacting the first end of
the first gate structure and the second end of the second
gate structure.
17. The device of claim 15, further comprising:
a first channel region, the first gate structure extending
along multiple sides of the first channel region; and
a second channel region, the second gate structure extend-
ing along multiple sides of the second channel region.
18. The device of claim 17, further comprising:
a first source/drain region adjacent the first channel
region;
a second source/drain region adjacent the second channel
region, the first source/drain region separated from the
second source/drain region; and
a gate spacer between the first source/drain region and the
first gate structure, the gate spacer disposed between
the second source/drain region and the second gate
structure.
19. The device of claim 17, wherein the first channel
region and the second channel region are parts of different
transistors.
20. The device of claim 15, wherein the first distance and
the second distance are each in a range of about 91 nm to
about 500 nm.

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